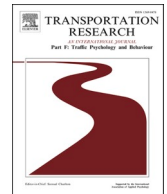




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Devil in the details: Systematic review of TOR signals in automated driving with a generic classification framework

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ABSTRACT

Background: Meta studies on factors contributing to take-over performance did not include the design of take-over request (TOR) signals, other than the modality at which TORs are presented. A detailed understanding of the influence of TOR design on take-over performance is therefore lacking.

Objective: To gain an overview of the level of detail with which TOR designs are reported in academic literature, by using and evaluating a novel classification framework. In this framework TORs are classified in terms of modalities, classes, and underlying attributes. Furthermore, the framework involves classification of potentially competing background signals, as well as the setting in which a study is performed.

Method: A systematic review was performed on articles written in English that were published between 2014 and 2021 using Web of Science, as well as articles retrieved from two previous TOR classification studies and three meta studies on take-over performance. Studies were considered for subsequent analysis if they involved a downward transition of the level of automation following a TOR, resulting in a sample of 391 TORs found in 189 studies.

Results: No predominant TOR design was found, and a considerable part of the available design space has not yet been explored. Studies reported less information on TOR designs when examining TOR designs at an increased level of detail. On average, attribute information was reported for half of the TORs per class.

Conclusions: More attention towards a detailed description of TOR implementations is needed and how this can impact experimental findings. The classification framework and the corresponding coding sheet could support systematic reporting and subsequent meta-analysis in future work. This way, a better understanding about the impact of TOR design on take-over performance can be gained, which in turn can support implementation of safe and effective TORs in (automated) vehicles.

1. Introduction

Developments in automobile technology have been ongoing ever since the first automobile was created. A large trend in recent years has been the effort towards higher levels of automation, with the ultimate goal of a vehicle capable of driving itself under all road

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conditions without intervention of a driver. Until that stage has been reached, drivers will at some point have to take over control of (parts of) the driving task in case automated technology has been activated. Whether caused by automation failure or by reaching the system's operational limits, transitions of control are typically signaled by a take-over request (TOR) as part of the vehicle's human-machine interface (HMI). For traffic safety, it is essential that drivers respond to a TOR by taking over whichever part of the driving task is required, in time (e.g., before a crash is imminent) and with good quality (e.g., appropriate situation awareness of the driver, limited variability in lane position of the vehicle). An important question for scientific researchers, automobile manufacturers and policy makers is how TOR design may optimize take-over performance. Recently, dedicated guidelines on HMI-design for automated vehicles have been introduced (Carsten & Martens, 2019; Naujoks et al., 2019b), which have been verified in terms of usability and user acceptance (e.g., Forster et al., 2020a), but an assessment of the effect of the quality of TOR design on traffic safety is missing.

Take-over time (TOT) and take-over quality (TOQ) following a TOR have been examined in several meta studies (Gold et al., 2018; Zhang et al., 2019; Weaver & DeLucia, 2022). Gold et al. (2018) modelled take-over performance in terms of TOT, time-to-collision (TTC), crash probability, and brake probability, based on six experiments involving SAE level 3 automation (SAE International, 2021) on a three-lane highway. Independent variables were time budget (i.e., available time to take over), traffic density, engagement in non-driving related tasks (NDRTs), repetition, initial lane, and driver age. The resulting models showed a strong influence of time budget, traffic density and repetition on take-over performance, whereas the other independent variables explained a minor portion of the variance in take-over performance. Zhang et al. (2019) focused exclusively on take-over time in their meta-analysis of 129 experimental studies with at least SAE level 2 automation. Out of a set of 18 independent variables, the researchers found that the mean take-over time decreased with a higher urgency of the situation, not using a handheld device, non-involvement in a visual NDRT, repetition of the scenario, and a unimodal auditory or vibrotactile TOR compared to a unimodal visual TOR or the absence of a TOR. Finally, Weaver and DeLucia (2022) performed a meta-analysis on 51 experiments with SAE level 3 automation, in which take-over performance was operationalized in terms of take-over time (i.e., gaze reaction time, road fixation time, hands-on wheel time, and time until the driver's first general input) as well as take-over quality (i.e., minimum distance, braking magnitude, steering magnitude, lane positioning, number of collisions). Time budget, NDRT engagement, and information support during the takeover (e.g., highlighting the cause of a TOR, recommending a specific response) were coded as independent variables. The analysis showed that a shorter time budget was associated with faster responses on the take-over time measures, but a worse take-over quality. Furthermore, NDRT engagement was associated with longer take-over timing and worse take-over quality. No significant effects of information support (e.g., highlight the cause of a TOR, recommend actions) on response time, nor take-over quality.

In the above meta-analyses the design of a TOR has been considered in terms of time budget (all studies), sensory modality (Zhang et al., 2019), and information support (Weaver & DeLucia, 2022). Other than that, in our view any variation in TOR design has essentially been treated as noise in the statistical analyses, despite various studies suggesting that (more detailed parameters in) TOR design could have an influence on take-over performance. For example, Vlakoveld et al. (2018) used a TOR consisting of an auditory alarm in combination with the words 'take over' on a display, and suggested that other TOR designs may have shortened the time required for situation awareness to develop. As another example, previous research showing how manipulating the time between pulses (e.g., heartbeats) has been used to successfully convey urgency (Van der Heiden et al., 2017; Van den Beukel et al., 2016).

If TOR design has not been subjected to statistical modeling at a more detailed level (thus treating details as noise), then the effect of detailed TOR design remains largely unknown, and therefore, there is no reason to believe that those parameters in TOR design have no effect. Gold et al. (2018) report corrected coefficients of determination of 0.43 and 0.63 for the continuous variables TOT and TTC, respectively. This implies that a large part of the variance in the data could not be explained by the independent variables. Zhang et al. (2019) found a strong effect of the study-specific error term, indicating that there are unobserved study variables that affect TOT; a limitation also mentioned by Weaver and DeLucia (2022). If the underlying studies prove to differ largely in terms of how TORs were implemented, then the question arises to what extent more detailed parameters in TOR design may have accounted for (some of the) the unexplained variance in the former meta-analyses.

To investigate the influence of TOR design on TOT and TOQ, one first needs to know if, and at which level of detail TOR designs are described in literature. For this reason, we have performed a systematic review on TOR designs, which was made possible by a novel classification framework postulated below.

1.1. Classification framework

Frameworks can generally be divided in frameworks that are focused on function and frameworks that are focused on form (Cabral et al., 2016). With a focus on function (e.g., Carsten & Martens, 2019), a framework details the purpose or activity of the human-machine interface (HMI), which a TOR is part of ("why"). Frameworks with a focus on form (e.g., Mirnig et al., 2017; Boelhouwer et al., 2019; Dey et al., 2020) detail the structure, manner or shape of the HMI ("how"). HMI functions may be described in terms of the process or the goals (e.g. inform, warn), whereas HMI forms may be described in terms of quantifiable properties for design and analysis (e.g. modality, size, intensity, location).

A framework for the assessment of specific TOR design choices in improving the automated driving task needs to allow for examining the potential of specific TOR forms. Previous form-based frameworks do not appear to provide enough level of detail to classify implementations of TOR designs encountered in literature. Take, for example, the study of Petermeijer et al. (2017c), who report an auditory TOR involving two single 2700 Hz beeps of 240 ms, spaced at 100 ms, and with a loudness of 80.6 dB. The classification frameworks of Mirnig et al. (2017) and Boelhouwer et al. (2019) do not include TOR frequency, duration, or intensity; properties which have been shown to influence perceivability, detectability as well as identifiability (e.g., Nees & Walker, 2011). Therefore, a framework to classify TOR design in more detail is needed, and below we describe why such details may be relevant for

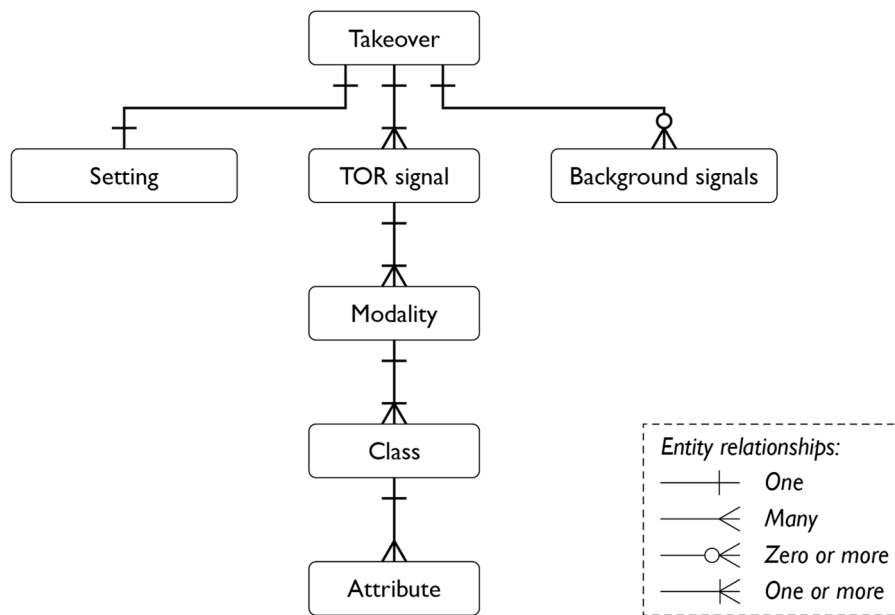


Fig. 1. Hierarchical framework for TOR signals within the context of a takeover situation. Connections between blocks follow crow foot notation (Everest, 1976). For example, a takeover situation involves one setting and one or more TOR signals, each of which are implemented using one or more modalities.

takeover performance.

Throughout the following sections, we postulate a form-based framework to classify TOR signals. A takeover situation may involve one or more TOR signals, each of which may be presented through one or more modalities. Within each modality (e.g., visual), one or more classes may be used (e.g., text, symbols), each of which can be described according to a series of attributes (e.g., phrase, color). Thus, TOR signals can be described at three hierarchical levels (see Fig. 1). In addition, background signals (e.g., music, involvement in a non-driving related task), if present, influence the extent with which TOR signals can be detected and discriminated. Finally, setting influences the response to a TOR (e.g., which controls are used to deactivate automation). The ensuing sections provide more details on the blocks depicted in Fig. 1.

1.1.1. Classification based on signal modality

Several frameworks take into account what specific sensory modality would serve a specific function. For example, Drüke et al. (2018) classify HMI design based on the urgency of warnings and interventions and indicate for each level of urgency the sensory modality that might be suitable for communicating about the warning/intervention to the driver. Mirnig et al. (2017) and Boelhouwer et al. (2019) categorize HMI design specifically for transitions to another automation level (i.e., following a TOR signal), including the sensory modality that is used to communicate on the transition. Likewise, Dey et al. (2020) include sensory modality in their taxonomy to classify external HMIs of automated vehicles. Furthermore, Naujoks et al. (2019b) developed a checklist to evaluate HMI designs, in which several checklist items pertain to specific sensory modalities. Thus the importance of the sensory modality through which communication takes place has been acknowledged in previous frameworks focusing on form, and moreover, as discussed above, has been shown to influence takeover time (Zhang et al., 2019).

A distinction is typically made between the visual, auditory and tactile modalities (Drüke et al., 2018; Mirnig et al., 2017), as these are the most commonly used modalities for HMIs in automated vehicles (Mirnig et al., 2017). Sometimes smell, taste and vestibular are also being included (Boelhouwer et al., 2019), yet communicating through these modalities is more challenging as perception through these modalities is less precise (Stein, 2012). Moreover, it has been demonstrated that it is not only important to consider unimodal signals (stimulating one of the sensory modalities) but also multisensory signals (stimulating multiple sensory modalities) in HMI design. Multisensory signals can be bimodal (e.g., auditory + visual) or even trimodal (e.g., auditory + visual + tactile). Research has shown that bimodal and trimodal signals can elicit faster responses to critical events during driving than unimodal ones (McNabb, 2017). Moreover, presenting a multisensory instead of an unimodal signal ensures that it is less problematic when a sensory modality of the driver is already taxed (Spence & Ho, 2008). For example, if the driver listens to loud music a signal that is both auditory and tactile will be less likely missed when presenting a signal that is only auditory. Moreover, the brain's susceptibility to unexpected signals is reduced during driving (Janssen et al., 2019). Therefore, especially for critical notifications it is essential to present a multisensory signal.

The present framework classifies the sensory modality of a TOR in terms of visual, auditory and tactile components, resulting in seven options: unimodal (visual, auditory, tactile), bimodal (visual + auditory, visual + tactile, auditory + tactile) and trimodal (visual + auditory + tactile).

1.1.2. Classification based on signal class

Besides the importance of signal modality and the number of sensory modalities stimulated in TOR design, there are specific signal classes (i.e., types of signals) within a modality that might affect the effectiveness of the communicated information. Previous frameworks (Mirnig et al., 2017; Boelhouwer et al., 2019) for example distinguish between acoustic (beep) sounds and speech sounds to distinguish between differences in length and informational content of speech versus non-speech auditory communication. Likewise, Naujoks et al. (2019b) indicate that complex auditory signals, such as speech output, can convey more detailed information than less complex auditory signals, such as beep sounds. Several signal classes that could be of importance in TOR design have been suggested in the literature and are discussed below for each modality.

Visual classes.

Text is regarded to be a specific type of visual signal (e.g., Dey et al., 2020; Montalvo et al., 2020). Text messages can be less ambiguous than other types of visual signals. Yet, shortcomings of text messages are for example that they are in a specific language and may not be universally understood and that they may not be readable at certain distances (Montalvo et al., 2020).

Anthropomorphic stimuli (using human-like elements for communication, such as facial expressions or gestures) are another specific visual signal class (e.g., Dey et al., 2020). Anthropomorphic representations in HMI design in automated vehicles, compared to non-anthropomorphic representations, have been shown to facilitate trust in and liking of an automated system (Niu et al., 2018). Moreover, Beller et al. (2013) showed that an anthropomorphic stimulus is effective in communicating on automation uncertainty. The benefit of anthropomorphic representations in HMI design is that it can enable untrained users to understand and predict the behavior of a machine (Niu et al., 2018). Yet, phenomena such as the uncanny valley (Mori et al., 2012), in which a human-like stimulus can appear almost human but slightly off and therefore elicits negative emotions, need to be avoided.

Symbols can also be considered as a specific signal class for the visual modality (e.g., Dey et al., 2020; Montalvo et al., 2020). Symbols are visual depictions (not containing human-like elements) that are recognizable and that convey a specific meaning (Dey et al., 2020; Rasmussen, 1983). Examples are an arrow indicating a direction and a red octagon symbolizing there is a need to stop.

Finally, an ‘abstract’ signal class is intended to capture signals that are neither textual, anthropomorphic or symbolic, such as LED bars or illuminated surfaces (following Dey et al., 2020). As described by Dey et al. (2020) these signals would communicate metaphors or these signals would be non-exact and therefore open to interpretation.

Auditory classes.

Speech signals are messages of the human voice, which can be pre-recorded or synthesized on the spot through text-to-speech algorithms. Nees and Walker (2011) argue that speech signals do not require learning (given that one is familiar with the language), but due to their length it may take relatively long to process the intended message. Speech has been used as distinct category in previous classification frameworks for TOR signals (Mirnig et al., 2019), external HMIs (Dey et al., 2020), as well as in guidelines for HMI design (Naujoks et al., 2019b).

Non-speech auditory signals have often been grouped, labeled as, e.g., ‘acoustic’ or ‘abstract’ signals (Mirnig et al., 2019; Dey et al., 2020). However, Nees and Walker (2011) identify several non-speech classes for in-vehicle technologies, each with unique strengths and weaknesses in terms of design flexibility and learnability.

Auditory icons are described as brief, often naturally occurring sounds that have some a priori association with their referent object, event, or process. Examples in the automotive domain range from car horns and skidding tires to signal an imminent collision to bicycle bells and footsteps to warn about the presence of vulnerable road users (Cabral & Remijn, 2019). Auditory icons generally yield faster reactions times and higher response accuracy compared to other non-speech signals described below (Cabral & Remijn, 2019), at times resulting in better performance than established standards (McDougall et al., 2020). Furthermore, they are relatively easily learned and yield a higher user acceptance. However, auditory icons can be misinterpreted when they share the same acoustic characteristics and they are more easily masked by environmental sounds (Cabral & Remijn, 2019).

The ubiquitous ‘beep’ sound is part of the ‘earcon’ class. Lacking an a priori association with its referent (as opposed to auditory icons), earcons can be considered as ‘abstract’ sounds. They often involve rhythms and/or melodies, based on which families of related sounds are created (e.g., two tones with an upward pitch interval to indicate system activation and a downward pitch interval to indicate system de-activation). Other examples fitting within this definition range from sinusoidal tones and frequency modulated tones to tones generated by musical instruments. Earcons are not prone to masking as long as the sound contains harmonics (i.e., no sinusoids), but are relatively difficult to learn and remember due to their abstract nature (Nees & Walker, 2011).

When an auditory signal represents quantitative data, it is called a ‘sonification’ (Nees & Walker). As such, properties of a warning sound are modulated by a phenomenon occurring outside the vehicle. The parking sensor is nowadays a common example: as one’s vehicle gets closer to a parked vehicle, pulses of the auditory signal speed up, an effect reported to increase perceived urgency (Van der Heiden et al., 2017; Van den Beukel et al., 2016).

Tactile classes.

Within signal classes for the tactile modality we can distinguish vibrational stimuli (Cabral et al., 2016; Kroon et al., 2019; Naujoks et al., 2019b) and force feedback stimuli (Adell et al., 2008; Yin et al., 2018). Vibrational stimuli are the most commonly applied type of tactile stimuli in HMI design (Naujoks et al., 2019b). Tactile vibrations can for example be embedded in the seatbelt or in the driver’s seat (e.g., McNabb, 2017; Petermeijer et al., 2017a). Intentional force feedback is sometimes used for communicating between the automation and the human driver. For example, Adell et al. (2008) applied a constant force on the accelerator pedal as a warning to support safe speed and safe distance.

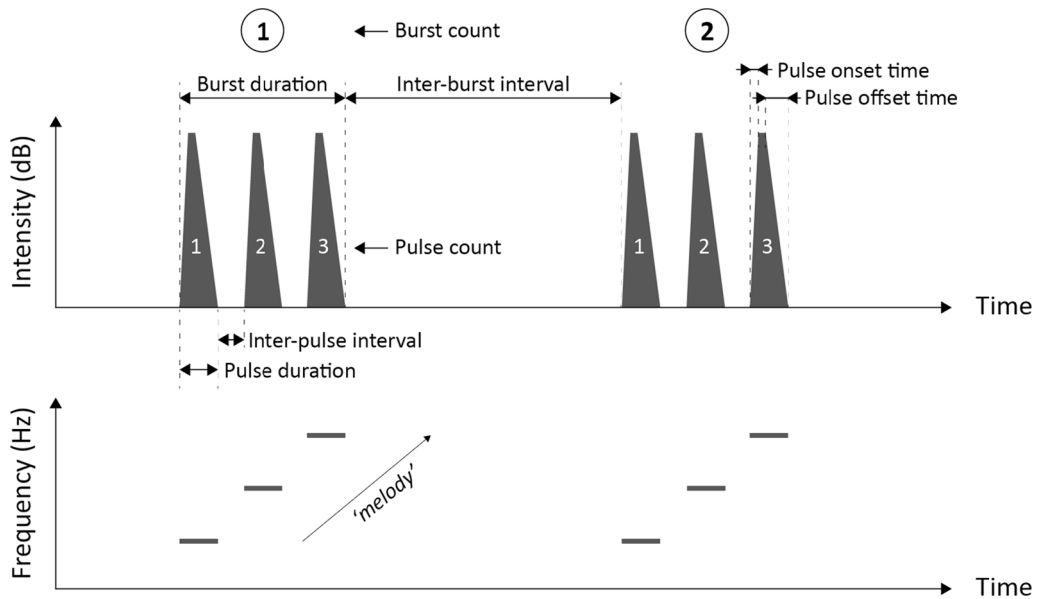


Fig. 2. Generic attributes relating to intensity and frequency over time. Depicted are two bursts of three pulses with a rising melody.

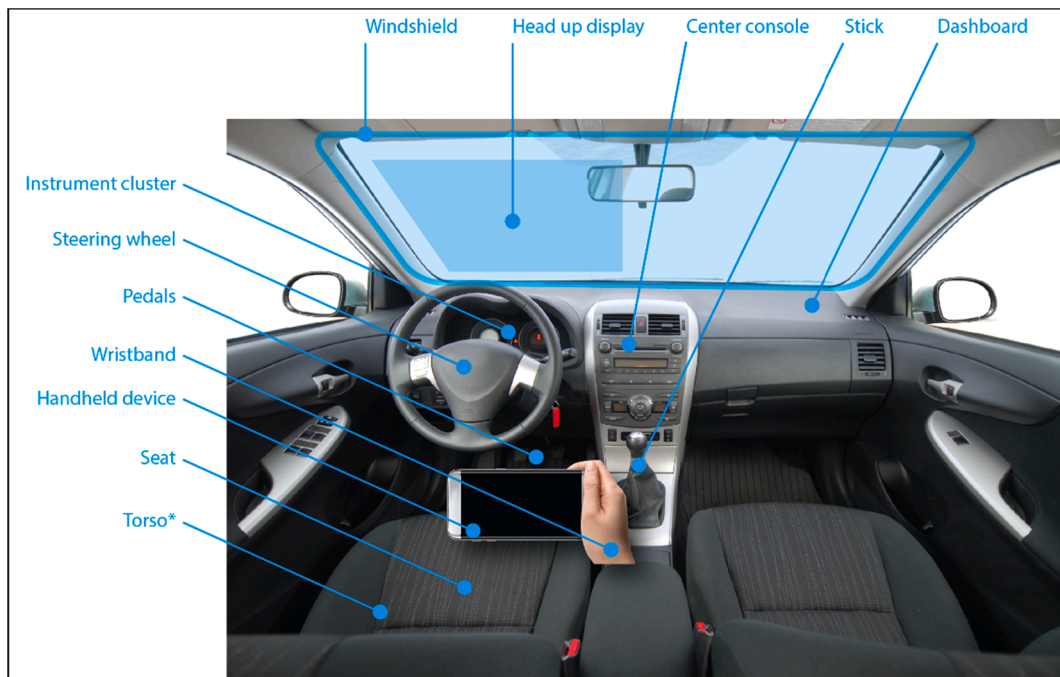


Fig. 3. List of locations where a TOR can be presented. Note: windshield includes the A-pillars as well as the edge above the window. * Torso refers to the torso of the driver (not depicted). Adapted from: Car Interior Stockphoto, In: iStock, 8 October 2013, Retrieved 10 June 2020, from: <https://www.istockphoto.com/nl/foto/car-interior-gm186160241-27615665>. Copyright by iStock. Adapted with permission.

1.1.3. Classification based on signal attributes

Each signal class can be described in terms of a series of attributes. Cabrall et al. (2016) even indicate that such a decomposition is necessary for comparing and establishing effects of various HMI implementations on the interaction between driver and automation. Geldard (1960) defined a set of four generic parameters to present information in tactile stimuli: locus, intensity, frequency, and duration. It appears that such parameters are applicable to classes in other modalities as well. For example, Petermeijer et al. (2017c) as well as Politis et al. (2017) present comparable signal implementations across the visual, auditory and tactile modalities, with regards to duration and inter-stimulus interval. Yet, there are also implementation properties specific for a signal modality or class. For

Table 1

Distribution of class-specific attributes across classes, indicated with an 'X'. Force feedback does not include class-specific attributes.

Attribute	Text	Symbol	Anthropomorphic	Abstract	Speech	Auditory icon	Earcon	Sonification	Vibrotactile	Force feedback
Material		X	X	X		X	X	X		
Phrase	X				X					
Syllable count	X				X					
Language	X				X					
Intonation					X					
Gender					X					
Font name	X									
Font style	X									
Font serif	X									
Width	X	X	X	X						
Height	X	X	X	X						
Distance to driver	X	X	X	X						
Size vibration area									X	
Sonification source								X		
Sonification destination								X		

example, visual text stimuli include properties such as font type and font style (Naujoks et al., 2019b) that are not applicable to any of the other signal classes. It therefore appears that both generic attributes (applicable to multiple classes) and class-specific attributes should be included in a form-based framework to classify HMI-design.

Generic attributes.

Generic attributes such as signal intensity, inter-pulse interval, onset time, offset time, pulse duration and the number of pulses are able to influence the urgency with which signals are perceived (Naujoks et al., 2019b; Van der Heiden et al., 2017; Van den Beukel et al., 2016; Marshall et al., 2007), as well as their detectability and identifiability (Cabrall et al., 2016; Nees & Walker, 2011). The guidelines of Naujoks et al. (2019b) appear to use the words 'burst' and 'pulse' interchangeably, and Nees & Walker (2011) give an example of 'pulsing bursts centered around 2500 Hz'. For consistent use of terminology, we propose the operationalization depicted in Fig. 2, in line with the definitions in Marshall et al. (2007). Each TOR signal consists of one or more bursts, where each burst has a certain duration and an inter-burst interval in case of multiple bursts. Individual bursts consist of one or more pulses, each with a certain duration, onset time, offset time, frequency, intensity, and an inter-pulse interval in case of multiple pulses.

When, for example, an earcon uses distinct frequencies per pulse, the corresponding TOR will be heard as a melody. In general, the notion of bursts and pulses is applicable to all modalities. Intensity may be interpreted in terms of vibration or force amplitude (tactile signals), loudness (auditory signals) or luminance (visual signals). Likewise, frequency may be interpreted as vibration frequency (vibrotactile signals), pitch (auditory signals), or color (visual signals). Some exceptions apply, which are discussed below for class-specific attributes.

Presentation location is another relevant attribute that applies to all classes. For instance, Politis et al. (2017) demonstrated that a signal presented in a location where the driver's attention is focused improved response speed, accuracy, and lateral deviation in a handover situation. Furthermore, varying the location of a signal can be employed to attract attention towards a specific direction (McNabb, 2017). Fig. 3 depicts the locations where a TOR signal can be presented. Here it should be noted that although Presentation location as a variable applies to all classes, the set of possible implementations varies between classes. For example, tactile stimuli are unlikely found in the window or on the dashboard, whereas visual stimuli are unlikely found in the seat. Furthermore, in case an experiment is performed using a table-top setup (e.g., with a game steering wheel, one or more screens, and a set of loudspeakers), the locations should be framed as were they part of an actual vehicle interior as depicted in Fig. 3.

Attributes related to timing and location are also very important for multisensory signals. The signals should occur approximately at the same time and approximately in the same location to ensure that signals that stimulate multiple senses are integrated into one percept; this is also called multisensory integration (Stein, 2012). Differences in timing and location can be examined by reporting the TOT and presentation location for each class.

Class-specific attributes.

Class-specific attributes are listed in Table 1. Material is applicable to six classes and describes the shape or sound of the corresponding signal. The set of implementations within Material varies between the classes Symbol (e.g., hands on a steering wheel, foot on a pedal, traffic sign, arrow), Anthropomorphic (e.g., facial expression), Abstract (e.g., circle, square, line, point), Auditory icon (e.g., skidding tires, car horn), and Earcon and Sonification (e.g., sinusoid or pure tone, beep, musical tone, frequency modulated tone). The Material attribute is not applicable to the classes Text and Speech, which by definition can only involve a Phrase (e.g., 'Take over', or a countdown timer '10, 9, 8, ..., 0'). Following the guidelines of Naujoks et al. (2019b), knowing which phrase has been used helps to determine whether national language was used (for convenience this can be coded as a separate Language attribute), whether simple, non-technical language was used, whether abbreviations were avoided, and whether the phrase refers to functionality of the automation. Likewise, syllable count can be used to assess whether Text and Speech messages are as short as possible. In conjunction with pulse (i.e., phrase) duration, one can determine the Speech rate (for auditory speech signals), an attribute affecting perceived urgency and commandingness (Bazilinsky & De Winter, 2017). Intonation concerns the pitch contour in Speech (e.g., rising, constant, falling),

which has been shown to affect perceived urgency (Ljungberg & Parmentier, 2012). The study by Bazilinskyy and De Winter (2017) suggests that synthesized female voices in a speech signal are easier to understand against a backdrop of noise than male voices. Also, gender of the voice message has been shown to influence the accuracy of responding to route instructions (Jonsson & Dahlbäck, 2011). According to Naujoks et al. (2019b), legibility of text signals is influenced by character spacing, the ratio between stroke width and character height, and the ratio between character width and height. Such properties can be captured through attributes Font type (e.g., Arial, Helvetica) and Font style (e.g., regular, light, condensed). Furthermore, Naujoks et al. (2019b) advice the use of sans-Serif fonts. A separate attribute 'Font serif' has been included in the framework, in case the Font name is unknown.

For visual classes, the combination of attributes Width, Height, and Distance to driver allow for determining the visual angle of the signal, which influences its legibility (Naujoks et al., 2019b). Ho et al. (2014) investigated the effect of dynamic tactile cues on brake response time and found that the use of three tactors (i.e., vibration devices) resulted in a faster brake response time compared to using one or two tactors. This finding indicates that a larger vibration area (a specific attribute for the Vibrotactile class) may influence driving performance following a TOR.

Sonification source and destination correspond with the physical phenomenon that causes a change in the sound (e.g., decreasing the distance to a parked vehicle) and the specific auditory attributes affected by the former phenomenon (e.g., reducing the inter-pulse interval). A special case of sonification is looming, where an approaching sound-emitting object is associated with an increase in sound intensity. Gray (2011) found that a looming auditory collision warning resulted in a faster brake response time compared to a non-looming version (i.e., with a static intensity). Moreover, the brake response time increased when the time-to-collision (TTC) signaled by the looming warning was changed from early (compared to the actual TTC) to veridical, to late.

Finally, no class-specific attributes have been introduced for force feedback TOR signals. This class can be described using the generic attributes.

1.1.4. Classification based on background signals

To assess the detectability and discriminability of a TOR, it is necessary to collect information relating to the TOR itself as well as the auditory, visual, and/or tactile background in which the TOR is presented. Detectability concerns whether one can notice that a signal occurred against a background of potentially competing signals. For auditory TORs, a pre-requisite is that the sound is not acoustically masked by other sounds inside the vehicle (such as engine noise). Sinusoid TORs are more easily masked than complex tones with more harmonic content. As a general rule, sounds should be between 15 dB and 30 dB above the background noise (Nees & Walker, 2011; Naujoks et al., 2019b). Likewise, vibrotactile signals should be between 15 dB and 20 dB above the detection threshold for the respective body region (Naujoks et al., 2019b). A visual counterpart of detectability is contrast of foreground and background in terms of luminance and color, for which guidelines are present in Naujoks et al. (2019b). If a signal can be detected, discriminability concerns whether that signal can be discriminated from other signals within and across in-vehicle systems. This can be achieved by avoiding multiple sounds with similar acoustic characteristics, as well as by temporally separating competing sounds by at least 300 ms (Nees & Walker, 2011). For visual signals, Naujoks et al. (2019b) recommend to spatially separate high-priority messages (e.g., TORs) from low-priority messages, but no recommendations on temporal separation are provided. An indication for the impact of one visual signal on subsequent visual signal interpretation may be found in psychophysics, where slow adaptation to intensity fluctuations in sensory system cells has been shown to last around 10000 ms (Baccus & Meister, 2002).

Aside from background noise and signals (visual, auditory, tactile) generated by in-vehicle systems, experimental studies often employ non-driving related tasks (NDRTs) which may involve signals affecting the detectability and/or discriminability of a TOR. Moreover, meta studies have shown that NDRT engagement negatively influences take-over timing and take-over quality (Zhang et al., 2019; Weaver & DeLucia, 2022). Potential masking effects resulting from NDRT-signals should be taken into account when investigating the effect of TOR design on takeover performance.

Based on the above guidelines and observations, the present framework accommodates an assessment of detectability and discriminability as follows. For TORs with a visual component, the presence of (NDRT-)signals is coded within the time window running from 10000 ms prior to the TOR onset to the moment at which the driver takes over or has to take over. We expect that signals presented and ending before this time window are unlikely to influence TOR detectability and discriminability. Spatial and temporal proximity are defined as the minimum distance and minimum time between any of these signals and the TOR signal, respectively. TORs with an auditory component are coded with the same properties as TORs with a visual component. In addition, the presence of stationary background noise (e.g., engine noise, road noise, wind and rain noise, music) and its intensity are included. Finally, the presence of tactile NDRT and background noise, as well as the intensity of background noise are coded for TORs with a tactile component.

1.1.5. Classification based on setting

The categorization framework of Mirmig et al. (2017) on control transition interfaces lists several take-over properties that cannot be attributed to the TOR signal itself (e.g., modality, class, attributes) or background signals:

- Direction: concerns whether a take-over (i.e., from system to driver) and/or a hand-over (i.e., from driver to system) takes place.
- Initiation: is the transition initiated by the user, system, or both?
- Involvement: does the vehicle ask the driver to conform that the requested control has been taken back?
- Completeness: are all driving and monitoring functions transferred at once, or incrementally, or partially?
- Fallback: system strategies in case a transition is unsuccessful (e.g., stopping in the ego lane or in another lane).
- Input modality concerns how a driver can respond to a TOR (e.g., physical, touch, speech, gestures).

- Mode awareness: features used by the vehicle to inform the driver about the system state before the TOR (e.g., text, symbols, colors, audio).

These properties may shape the way drivers respond to TORs (and influence how designers implement TORs) and are therefore labelled as ‘setting’ in the present framework.

Higher levels of automated driving give rise to new human factor issues (Carsten & Martens, 2019) and therefore it is important to consider the specific automated driving level(s) that are activated during communication of a signal when assessing the HMI's effectiveness. This can be interpreted in more ways than what has been done in previous meta studies (Gold et al., 2018; Zhang et al., 2019; Weaver & DeLucia, 2022), which seem to focus solely on transitions from either partial, conditional, or highly automated (i.e., SAE level 2 and above) to manual driving (i.e., SAE level 0). Lu et al. (2016) postulate a driver-automation system with six states, each with a distinct distribution of responsibilities between the driver and automation in terms of three primary driving tasks: lateral control, longitudinal control, and monitoring. Furthermore, Lu et al. (2016) discern between two transition types between the states of the driver-automation system. First, ‘control transitions’ are transitions in which the execution of at least one control task changes between driver and automation (e.g., from level 3 to level 0 when interpreted in terms of the SAE levels of automation). Second, ‘monitoring transitions’ are transitions in which only the required monitoring frequency changes between driver and automation (e.g., from level 4 to level 2). A transition can take place in either direction, whereas a take-over has been framed as a transition from a high to a low level of automation, the latter typically manual driving (Mirnig et al., 2017; Zhang et al., 2019). Lu et al. (2016) state that a take-over may be initiated by an exceedance of the automation's operational limits. Take-overs have mostly been studied as control transitions, but the exceedance of the automation's operational limits could also refer to its monitoring abilities in a particular context. Likewise, take-over requests (TORs), which have been used mostly for control transitions, may also be used to signal monitoring transitions (e.g., Forster et al., 2020b). Furthermore, several measures relevant for studying control transitions are likely also relevant for monitoring transitions, such as gaze reaction time and hands-on-wheel time. The present framework therefore includes TORs for control transitions, as well as monitoring transitions.

Setting also involves properties situated outside the vehicle. Previous meta studies show that traffic density influences take-over timing and take-over quality (Gold et al., 2018; Zhang et al., 2019). Therefore, traffic density is included in the present framework such that the effect of traffic density can be controlled for in a future meta study on the effect of TOR design on take-over performance.

1.2. Study objective

The classification framework presented above serves two aims: 1) to support systematic reporting of TOR design parameters in future studies, and 2) to support inventorying of TOR design parameters used in previous studies. The framework includes a hierarchical classification of TOR signals at three levels: 1) modality or modalities through which the signal is presented, 2) classes within a modality in which the signal falls, and 3) description of those classes according to attributes. Additionally, the framework considers background signals and the experimental context which is thought to shape the interaction with a TOR signal. The framework is a valuable addition to previous work that mostly considered (more detailed) TOR design as noise in statistical analyses. The impact of TOR design on take-over performance is currently unclear. Moreover, if one wants to gain insight into the effects of TOR design the design space should be explored broadly. It is currently also unclear how much of the design space has been employed in previous studies.

A logical next step is to apply the proposed framework in order to create an inventory of TOR implementations in previous studies. The goal of such an inventory is to gain insight into the utilized design space and, ultimately, to establish effects of specific aspects of TOR design through future meta-analyses. An important prerequisite to reach these goals is that details considered in the framework are actually reported in previous work. Therefore, first it needs to be established to what degree implementation of TOR design can be deduced from previous research in order to determine whether insights can be gained into the utilized design space and whether meta-analyses will be meaningful. We next present a systematic review in which the framework has been used to categorize previous experimental studies.

2. Method

A systematic literature search was performed using the framework postulated above, to examine at which level of detail TOR designs have been reported in literature. This study is reported conforming to the Preferred Reporting Items for Systematic Reviews and meta-Analyses (PRISMA) guidelines (Moher et al., 2009). The PRISMA checklist is available in Appendix A. No protocol was generated or registered.

2.1. Information sources & search strategy

Studies used in the frameworks of Mirnig et al. (2017) and Boelhouwer et al. (2019), as well as the meta-analyses of Gold et al. (2018), Zhang et al. (2019), and Weaver and DeLucia (2022) were selected for subsequent screening (see below). In addition, a literature search was performed in Web of Science using a query with two keyword groups in the topic field. Group one consisted of ‘selfdriving’, ‘autonomous’, ‘automated’, ‘self-driving’, and ‘self driving’. Group two consisted of ‘handover’, ‘hand-over’, ‘hand over’, ‘takeover’, ‘take-over’, ‘take over’, and ‘transition’. Keywords within each keyword group were connected with logical ‘OR’ operators, whereas the keyword groups were connected with a logical ‘AND’ operator. The language of the query was set to English, including

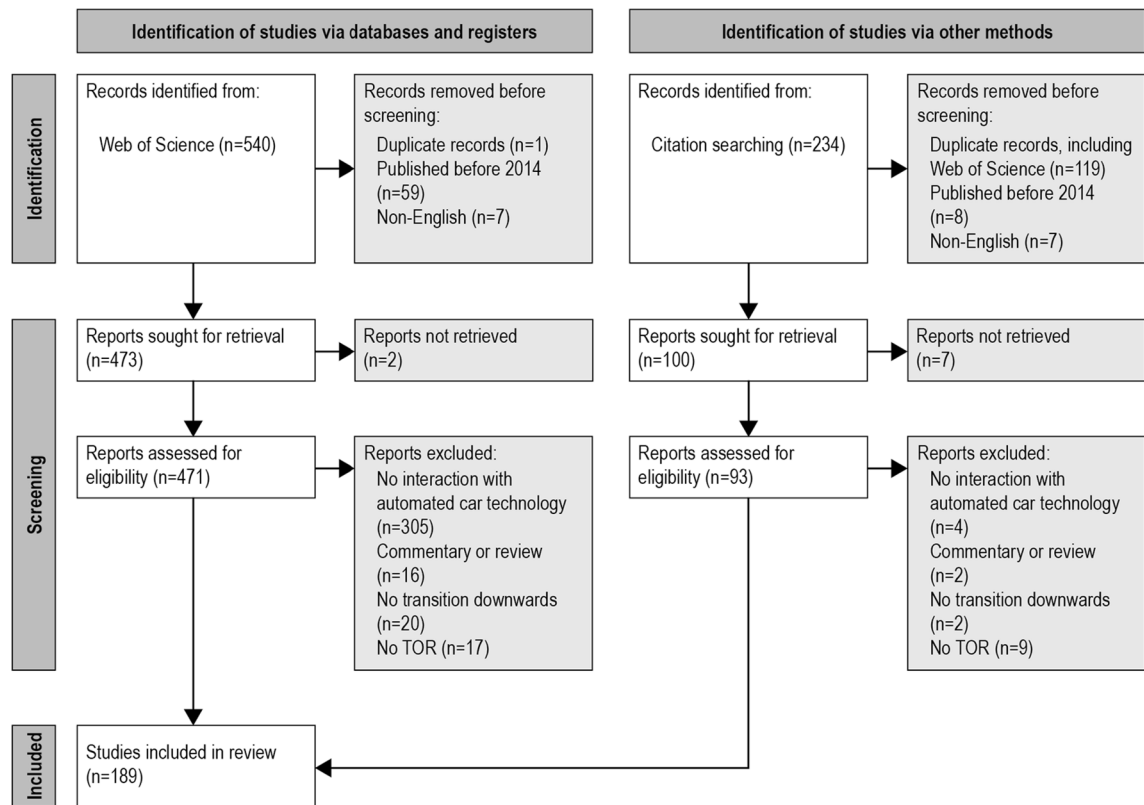


Fig. 4. Data reduction resulting in the final set of included studies.

articles, early access, as well as proceedings papers for publications between 2014 and 2021. Included indexes were SCI expanded, SSCI, A&HCI, BKCI-S, BKCI-SSH, and ESCI.

We deliberately chose to include proceedings papers to yield an as large as possible sample, but acknowledge that especially proceedings papers may suffer from selective reporting of TOR implementation details due to restrictions on the manuscript length. The query was performed on March 30th 2021. Together with the papers derived from the meta-analyses, a total initial sample of 719 papers was subjected to the screening process described in the next sections.

Production cars were not considered in the present review. [Mirnig et al. \(2017\)](#) note that it is difficult to keep track of the exact implementation of signals related to autonomous driving technologies, due to rapid changes in their development. Also, today's production cars are not (yet) legally allowed to use automation beyond level 2, and cars with level 3 automation have been introduced only recently (e.g., [Templeton, 2021](#)), which means very few examples of TOR signals are expected. Industry patents were deliberately excluded, too. Patents do not require a fully working implementation, and may cover several different implementation concepts to secure a market advantage ([Mirnig et al., 2017](#)). This reduces the likelihood of unambiguously coding the implementation of TORs covered by industry patents; a necessity in the present review.

2.2. Eligibility criteria

Studies had to fulfill the following five criteria to be included in this review:

1. The study had to involve a control transition or monitoring transition from partially, conditionally, or highly automated driving (i.e., SAE levels of automation 2–4; hands off the steering wheel and feet off the pedals) to any lower level of automation (i.e., SAE levels of automation 0–3).
2. The study had to involve a transition in response to a 'take-over request' (TOR). That is, we included only 'mandatory automation initiates driver-in-control (AIDC) transitions' as defined by [Lu et al. \(2016\)](#).
3. The TOR had to be implemented physically, as opposed to a hypothetical TOR that had to be imagined (e.g., [Bazilinskyy & De Winter, 2015](#)).
4. Excluded were non-English language studies, studies that did not describe automation technology in cars, reviews, commentaries, and studies of which the full article was not accessible.
5. Following [Mirnig et al. \(2017\)](#), only publications starting from 2014 were considered, as this year coincides with the first publication of the NHTSA (late 2013) and SAE (early 2014) levels of driving automation.

2.3. Study selection & data extraction

Identification of 774 records resulted in a final set of 189 included studies, as follows (see Fig. 4). First, duplicate records were removed from the initial sample identified through database and literature search, based on the combination of authors, manuscript title and publication year. Based on the reference and additional details provided by Web of Science, records which proved to be non-English, and/or published before 2014 were removed. The remaining records were distributed across the four authors to verify whether a report met the eligibility criteria (i.e., each report was screened by one author). Screening started with the title and abstract. The full paper was assessed in case not all eligibility criteria could be verified. When a report proved to disagree with any of the eligibility criteria listed in the previous section, the report was not further screened on the remaining eligibility criteria. No steps were taken to avoid double counting due to different reports of the same study (e.g., published in another year under another title). Thus, each report has been treated as a unique study. This approach was chosen to provide the best possible insight into the deducibility of TOR design parameters as they are encountered in individual publications.

A codebook (see Appendix B) was developed based on the classification framework presented in Section 1.1. The codebook was operationalized in Microsoft Excel (v2203), offering separate sheets for instructions, background signals and setting, as well as seven separate sheets for all combinations of the visual, auditory and tactile TOR modalities. Within each of those seven sheets maximally-four distinct TORs could be coded. This setup was based on an initial survey of studies used in the meta-analysis of Zhang et al. (2019), where Politis et al. (2017) proved to be the study with the largest number of distinct TORs: 28 in total; 4 TORs per modality sheet. The operationalized codebook was pilot-tested on two studies randomly selected from the sample, and refined accordingly. Studies were coded in-depth by authors RJJ, SvdK, and RdZ. Only information reported within a study was considered; corresponding authors were not contacted and no snowballing techniques were used to derive additional information in case references to previous experimental setups were made. However, note was taken of studies providing such references.

2.4. Analysis methods

Analysis of the coded papers has been performed for each of the blocks in the hierarchical framework depicted in Fig. 1. From this framework, it follows that TOR classes can only be examined when the corresponding TOR modality is implemented. Likewise, TOR attributes can only be examined when the corresponding TOR class is implemented. As will be shown, a key observation of our analysis is that variables in the codebook were often found unreported in the studies under scrutiny. To provide insight into the number of missing values in the following analysis, we report the proportion of ‘deducible’ values for each variable in the codebook. Deducibility here refers to whether a study provides any information about a variable (e.g., a unimodal visual TOR was used), including information based on which the absence of an implementation could be determined (e.g., the TOR did not involve an auditory component). It follows that deducibility of a variable (e.g., a TOR phrase in case text is used) involves the entire sample (i.e., including TORs in which the phrase is coded as ‘unknown’). When reporting proportions of ‘known’ categories within a variable, however, cells coded as ‘unknown’ are left out of the equation. In other words, other than deducibility, reported counts and proportions regarding TOR implementation represent only the ‘known’ subset of the sample.

A dedicated R-script was written to process the Excel sheets with coded studies, to show the coders which entries were overlooked, and ultimately to verify that all applicable fields were entered. The base package (version 4.1.1) was used to calculate proportions, means and standard deviations. Package ‘ggplot2’ (version 3.3.5) was used for visualizations.

In terms of implementation of TOR attributes (which turned out to show a relatively low deducibility), we focus on attributes of which a value is known in at least 50 % of the corresponding class. Furthermore, we distinguish between attributes with low variability (3 or fewer unique categories, or a dominant category) and attributes with high variability (>3 unique categories without a dominant category) to describe the range of the design space used. The rationale for these thresholds is that if an attribute with a deducibility of 50 % involves a single unique category or value, then a theoretical deducibility of 100 % would result in at least one dominant category. Conversely, if an attribute with a deducibility of 50 % involves a large number of unique categories without a dominant category, then a theoretical deducibility of 100 % unlikely results in one or two dominant categories. Thus, using a criterion of 50 % deducibility yields insights in the potential variability in the full TOR population, which in turn may inform the number of categories required in future meta studies.

Interrater reliability was assessed before and after coding of the studies, and is described in more detail in Appendix D.

3. Results & discussion

3.1. Sample description

A total of 391 distinct TORs were coded within the sample of 189 studies (see Appendix C). Studies with more than one TOR were due to manipulations of the TOR implementation (e.g., Politis et al., 2017) or because multiple TORs were presented in a single takeover situation (e.g., Forster et al., 2017). The majority of studies did not refer to other studies when describing a TOR implementation. Of the 27 studies with such references, all of which were included in the final sample, 18 studies (jointly accounting for 92 distinct TORs) involved an ambiguous reference, in that the TOR implementation was reported as an adjustment from earlier work (e.g., Politis et al., 2015), based on earlier work (e.g., Stanton et al., 2022) or guidelines (e.g., Wan & Wu, 2018a), or in line with previous work (e.g., Borojeni et al., 2016). The other 9 studies (involving a total of 20 TORs) explicitly stated that the TOR implementation was a copy from earlier work (Müller et al., 2021; Petermeijer et al., 2017c; Naujoks et al., 2014; Sportillo et al., 2018) or that more details

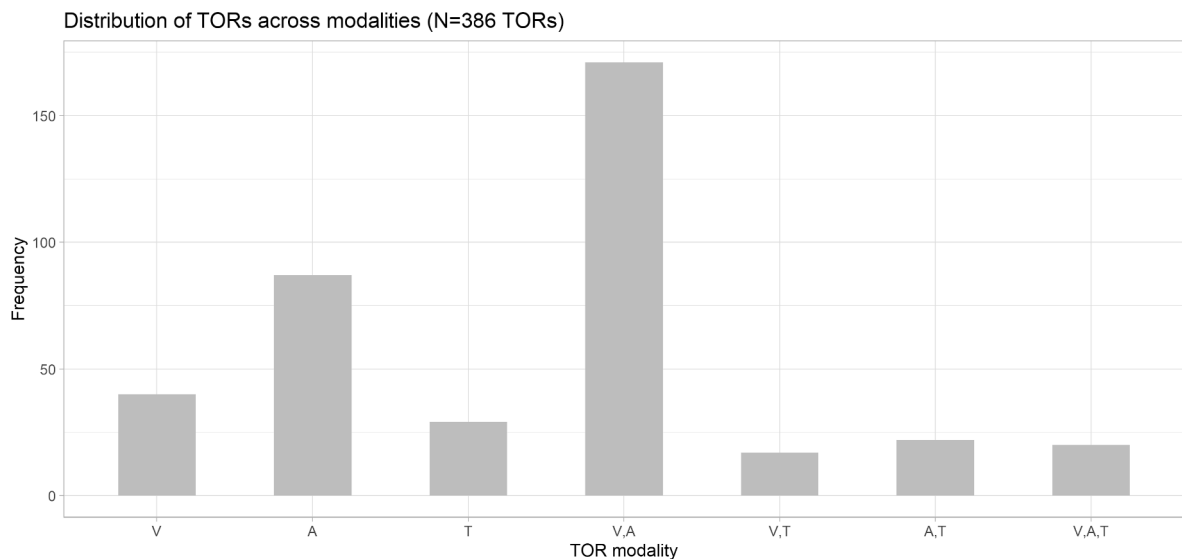


Fig. 5. Frequency of deducible unimodal and multimodal TOR implementations in 184 studies. Note: a single study may involve multiple TORs in distinct modalities. V = Visual, A = Auditory, T = Tactile.

about the TOR implementation could be found in earlier work (Forster et al., 2016; Forster et al., 2017; Körber et al., 2016b; Naujoks et al., 2017b; Wiedemann et al., 2018).

The coding scheme constructed for the present study allowed for a maximum of 4 distinct TORs per modality. In one paper (Chen et al., 2020) there were 5 settings for the takeover time (TOT), which required coding 5 TORs. In this case one TOT-setting was omitted. Another paper (Duthoit et al., 2017) tested 15 distinct tactile TORs, of which only four could be coded. Thus, a total of 12 TORs were identified, but not coded. For both papers, care was taken to select TORs that would cover the widest area of the design space. Furthermore, the coding scheme allowed for a maximum of 4 distinct frequencies (i.e., one per pulse) so that, e.g., short melodies can be included. One paper described a TOR with 8 pulses (Borojeni et al., 2018b). Seeing that the same frequency was used 8 times, the first four frequencies were coded and the remaining ones were omitted.

3.2. TOR modalities

3.2.1. Deducibility

Five studies stated that a TOR was presented to the driver without providing details on the modality of the TOR (Scharfe et al., 2020; Lee et al., 2021; Jarosch et al., 2017; Ebinali et al., 2019; Körber et al., 2016b). In two of these studies, a time budget was specified, but this could not be coded as attribute lacking information about modality and class. The modality of each TOR was deducible in the remaining 184 studies, resulting in a deducibility of modality in 386 (or 98.7 % of the) TORs. Note that a high deducibility here involves both unambiguous presence and absence of a modality.

3.2.2. Implementation

A total of 386 TORs were coded on modality. Fig. 5 shows how TORs were distributed across the modalities. All unimodal and multimodal TORs were represented in the sample of studies. A multimodal visual-auditory TOR was by far the most common implementation (44.3 %), followed by a unimodal auditory TOR (22.5 %). This finding is in line with Boelhouwer et al. (2019), where visual-auditory and auditory TORs were the largest groups, too, constituting 35.5 % and 19.4 % of the total number of TORs, respectively. In contrast, the largest TOR groups in Mirnig et al. (2017) were visual-auditory (42.9 %) and visual (22.9 %), with auditory TORs accounting for a mere 5.7 % of the sample. Thus, recent studies may have more often employed auditory TORs.

3.3. TOR classes

3.3.1. Deducibility

Several studies reported the modality of a TOR, but failed to indicate which class was implemented. Consequently, the deducibility of class ranged between 84.3 % and 89.5 % across the classes ($M = 87.1$, $SD = 1.9$). In some cases, lack of information on class meant that additional information at the attribute level could not be coded, for example: “Sounds are also emitted, especially-one for takeover, that lasts during the whole transition phase and is designed to sound more and more urgent.” (Langlois et al., 2016).

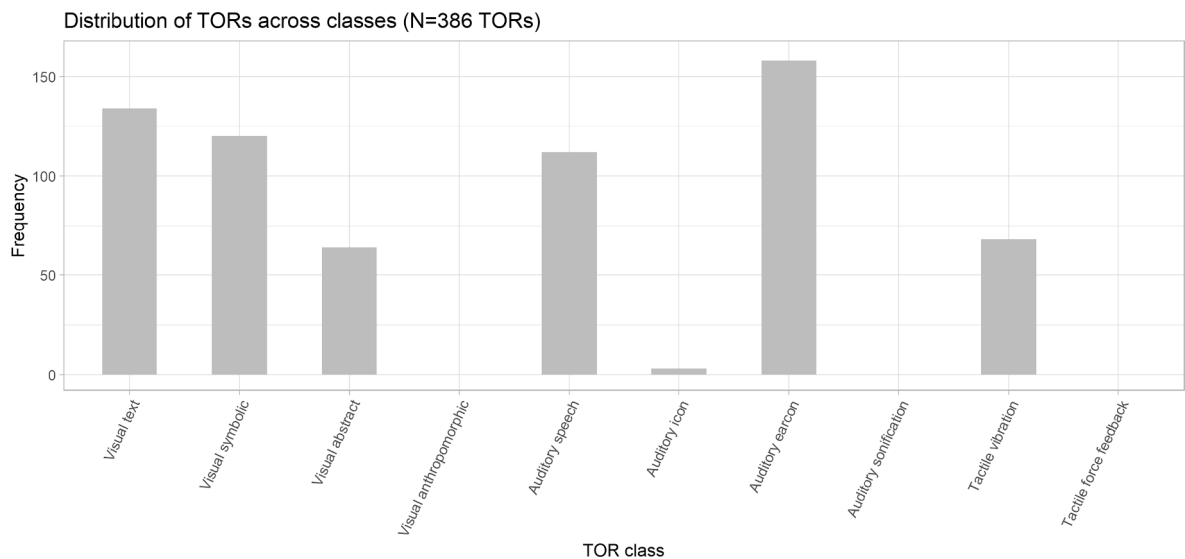


Fig. 6. Frequency of implemented TOR classes within deducible TOR modalities. Note: TOR classes may be implemented in unimodal as well as multimodal TORs.

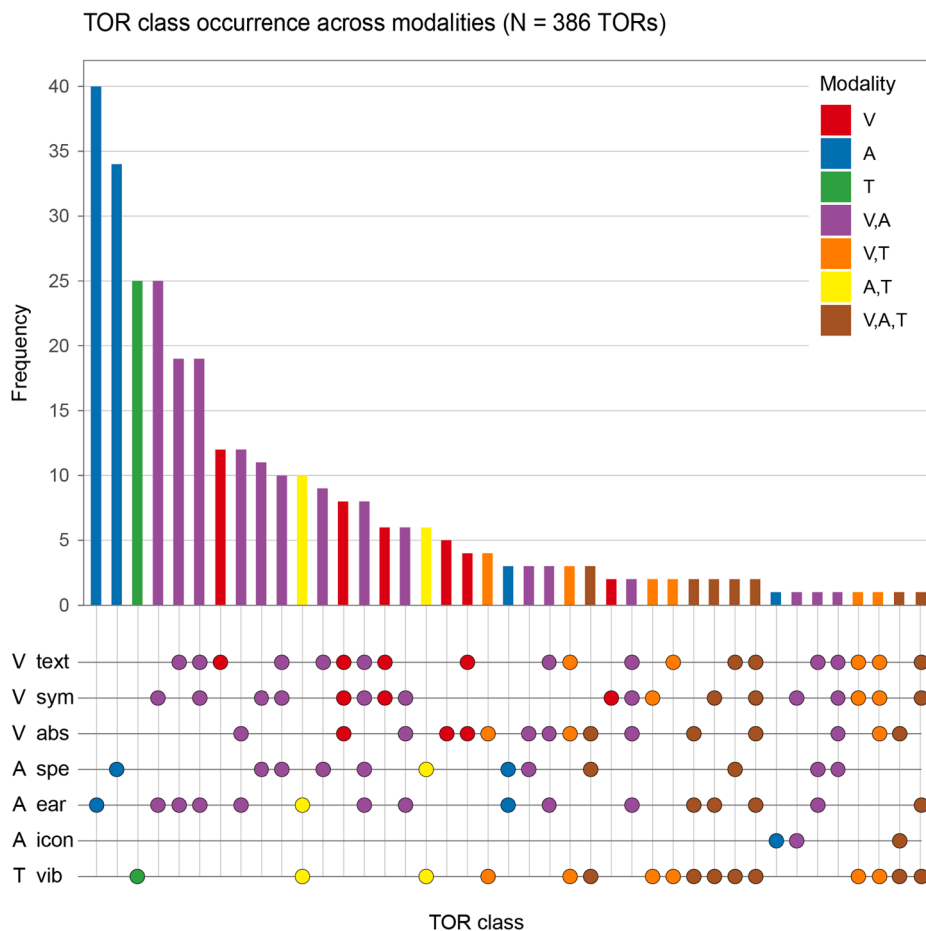


Fig. 7. Frequency of implemented TOR class combinations (indicated by circles) within deducible TOR modalities, calculated per TOR modality. V = Visual, A = Auditory, T = Tactile, abs = abstract, ear = earcon, icon = auditory icon, spe = speech, sym = symbol, vib = vibration. Note: TORs in which at least one class could not be deduced were excluded, because such missing data does not allow for identification of class combinations.

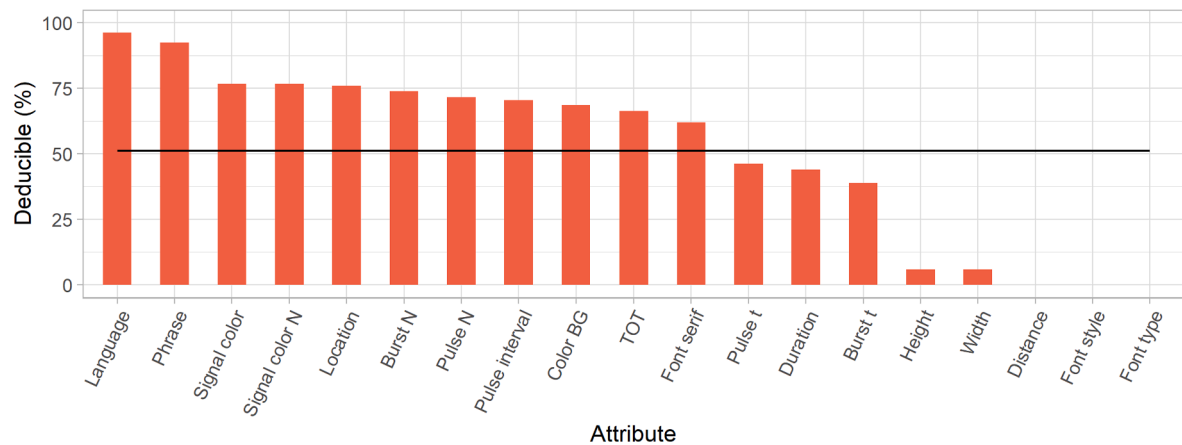
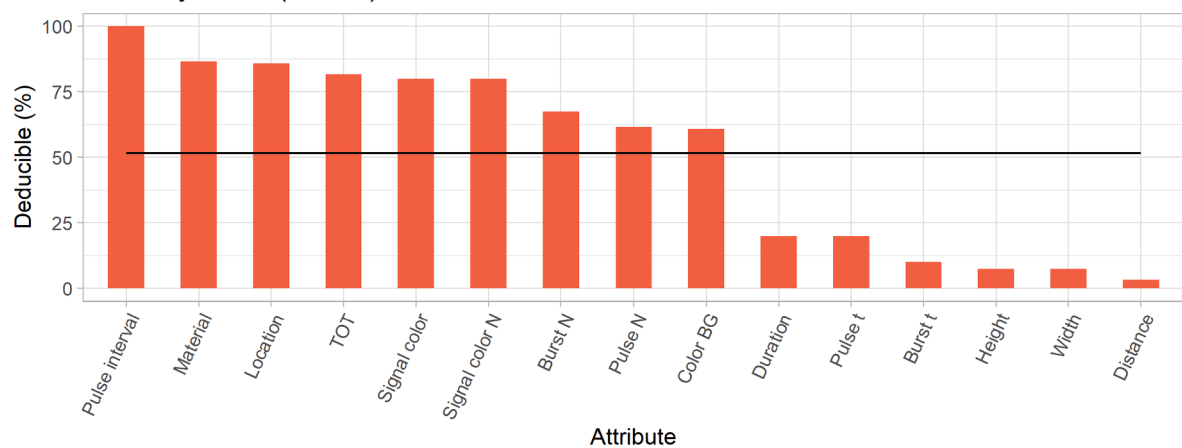
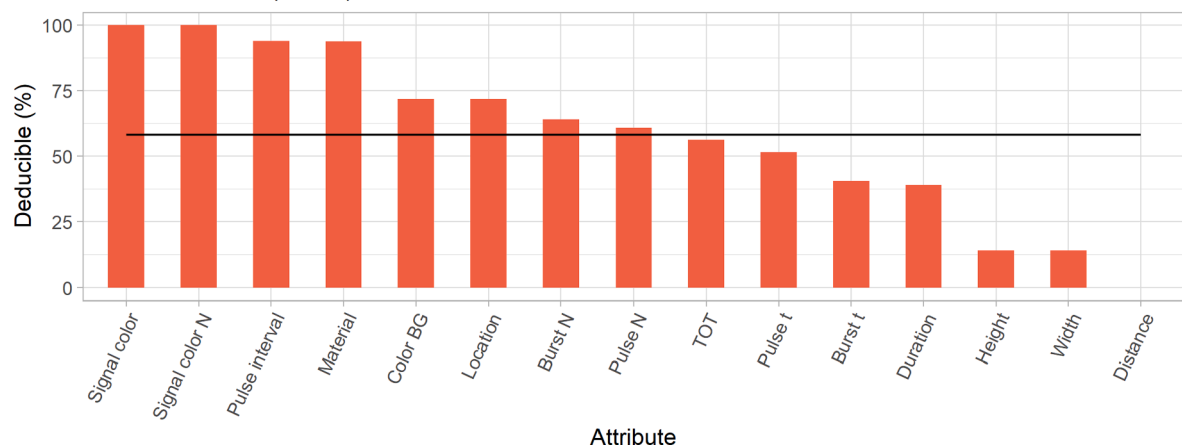
a Visual text (N=134)**b** Visual symbolic (N=120)**c** Visual abstract (N=64)

Fig. 8. Deducibility of TOR attributes for classes Visual text (a), Visual symbolic (b), and Visual abstract (c). Black lines represent the mean deducibility across attributes within a class.

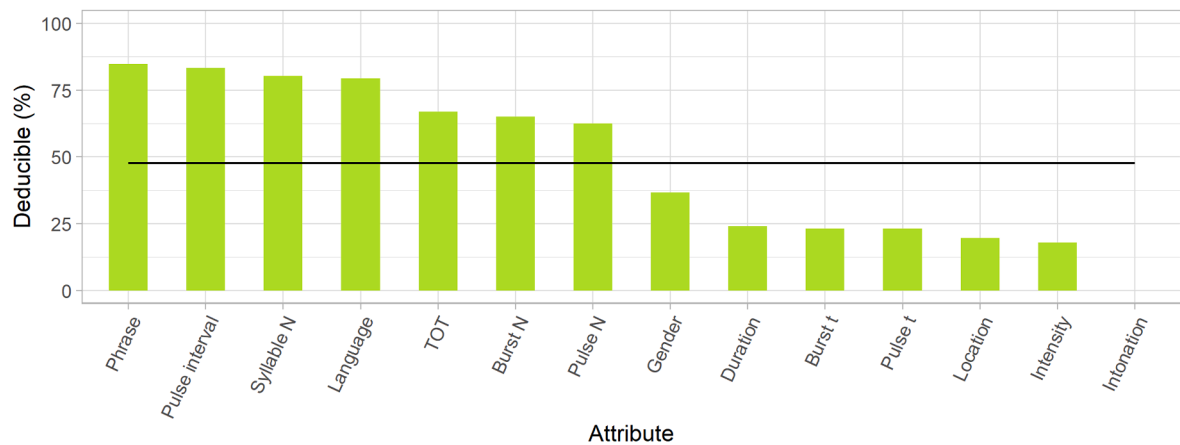
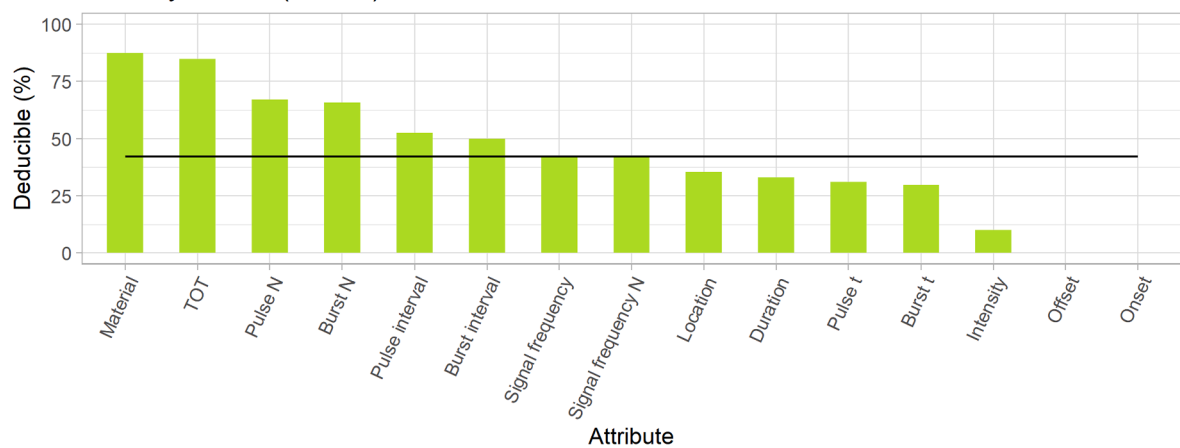
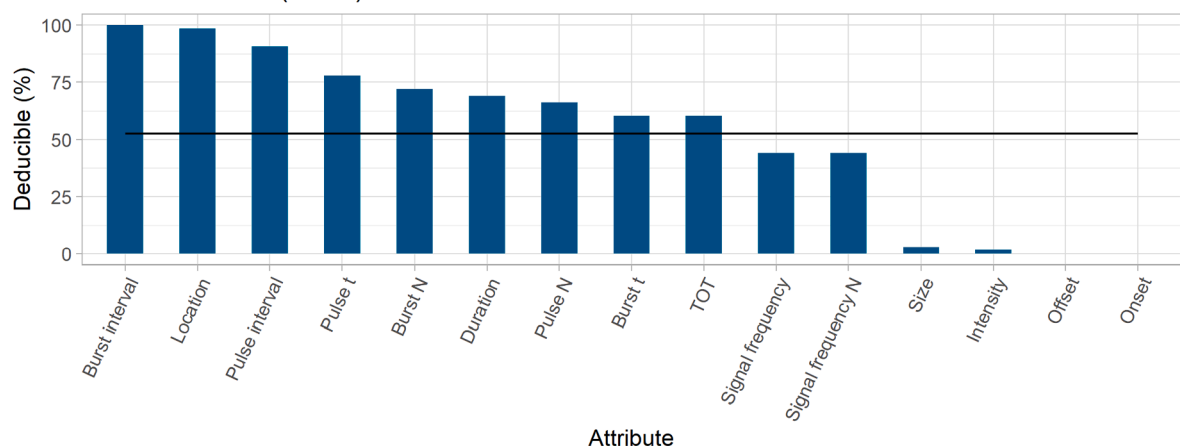
a Auditory speech (N=112)**b** Auditory earcon (N=158)**c** Tactile vibration (N=68)

Fig. 9. Deducibility of TOR attributes for classes Auditory speech (a), Auditory earcon (b), and Tactile vibration (c). Black lines represent the mean deducibility across attributes within a class.

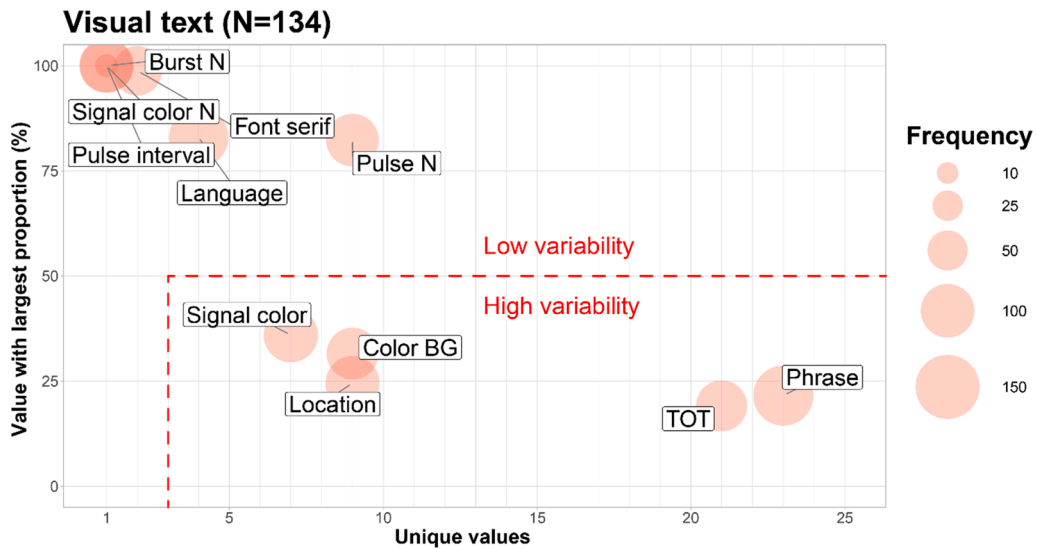


Fig. 10. Number of unique values and proportion of the largest value within attributes of the visual text class. A high variability is defined as attributes with at least three unique values and where the value with the highest proportion is represented in fewer than 50 % of the TORs. Only attributes with a deducibility of at least 50 % are depicted. ‘N’ = count, ‘BG’ = background, ‘TOT’ = takeover time.

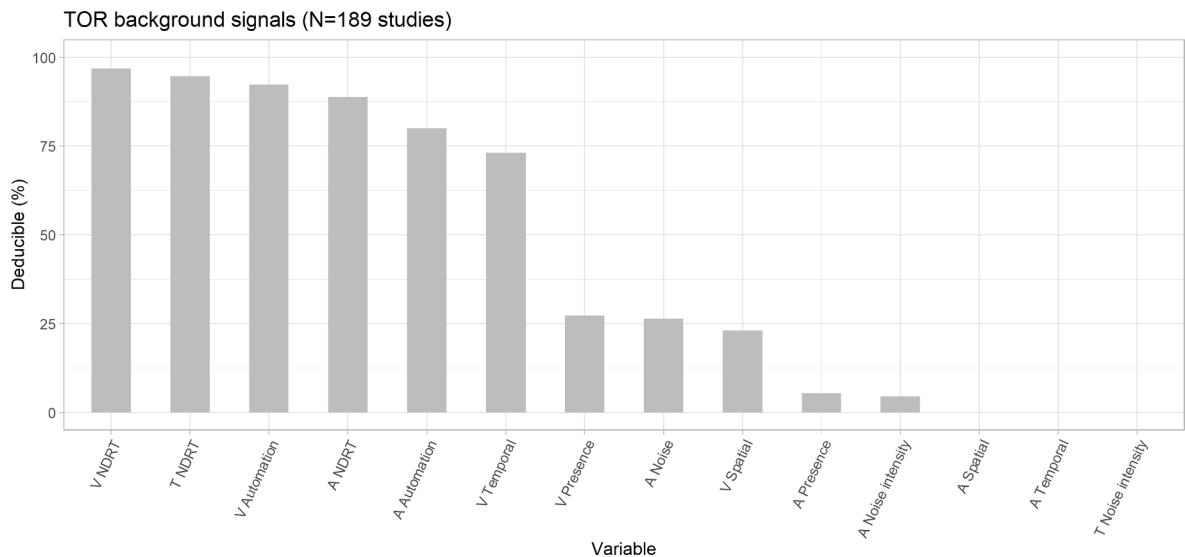


Fig. 11. Deducibility of background signals. V = Visual, A = Auditory, T = Tactile. Note: aside from NDRT-involvement, background signals were only coded if the corresponding modality was known.

3.4. Implementation

Looking at the actual implementation of TOR classes in Fig. 6, text and symbols are most often used in TORs with a visual component, whereas earcons (and to a lesser extent speech) are mostly used in TORs with an auditory component. Visual anthropomorphic, auditory sonification, and tactile force feedback classes were not observed in the sample. A comparison with previous studies classifying TOR designs (Mirnig et al., 2017; Boelhoutwer et al., 2019) cannot be made regarding these classes, because they were not included in the corresponding frameworks.

TORs may involve multiple classes. We observed 41 distinct implementations of (combinations of) classes in the sample. Fig. 7 depicts the frequency of class implementations per modality. For unimodal auditory as well as bimodal auditory-tactile TORs, earcons and speech are the most occurring classes, yet they are seldomly used together, in line with findings by Boelhoutwer et al. (2019). The earcon-speech combination is found in 4 unimodal auditory TORs (accounting for 3.8 % of the modality) and in 8 bimodal visual-auditory TORs (6.9 %). For unimodal visual TORs text is most often used exclusively (25 TORs; 32.4 % of the visual modality),

Table 2

High or low attribute variability across classes. Empty cells are the result of low deducibility or when attributes are not featured in a class.

Attribute	Class					
	Text	Symbol	Abstract	Speech	Earcon	Vibration
Burst interval					Low	Low
Burst count	Low	Low	Low	Low	Low	Low
Burst duration						High
Color background	High	High	Low			
Duration						High
Font serif	Low					
Language	Low			Low		
Location	High	High	High			Low
Material		Low	High		Low	
Phrase	High			High		
Pulse interval	Low	Low	High	High	Low	High
Pulse count	Low	Low	Low	Low	High	High
Pulse duration			High			High
Signal color	High	High	High			
Signal color count	Low	High	Low			
Syllable count				High		
Takeover time	High	High	High	High	High	High

followed by a combination with symbols and abstract shapes (8 TORs; 21.6 %). The bimodal visual-auditory TORs and the trimodal visual-auditory-tactile TORs have been implemented in a large number of combinations of text, symbols, abstract shapes, speech, earcons, and vibrations, and consequently show frequencies ranging between 1 and 25 TORs. All in all, the most frequently occurring modality (i.e., bimodal visual-auditory) does not show a predominant implementation in terms of classes, whereas the second most frequently occurring modality (i.e., unimodal auditory) features two common implementations, involving either speech or earcons.

3.5. TOR attributes

3.5.1. Deducibility

Several studies mentioned the modality and class of a TOR, but did not specify the underlying attributes, such as what type of sound was used, as well as its frequency, intensity and duration (e.g., [Vogelpohl et al., 2020](#); [Vlakveld et al., 2018](#)). [Fig. 8](#) displays the deducibility of attributes in three visual classes, regardless of the modality in which the class was used (e.g., unimodal, bimodal, trimodal). Likewise, [Fig. 9](#) displays the deducibility of attributes for the classes speech, earcons and vibration. Auditory icons have been excluded from subsequent analysis at the attribute level, because there were only 3 instances, all reported by a single study ([van den Beukel et al., 2016](#)). The mean deducibility across attributes within a class ranged between 42.1 % (auditory earcons) and 58.1 % (visual abstract), with a grand mean of $M = 50.5\%$ ($SD = 5.3$) across all classes presented in [Fig. 8](#) and [Fig. 9](#).

Temporal attributes have been coded in all classes. In terms of the signal repetition (i.e., burst count, pulse count), the deducibility ranged between 60.9 % and 73.9 % ($M = 66.4\%$, $SD = 4.1$) for all classes shown in [Fig. 10](#) and [Fig. 11](#). Inter-burst interval and inter-pulse interval were coded in case at least two bursts or pulses were reported, yielding a deducibility above 70.6 % for all classes, except for earcons (50.0 % and 52.6 %, respectively). For example, [Geitner et al. \(2019\)](#) report “a 2 s beep [...] pulsing seven times”, and [van der Heiden et al. \(2017\)](#) report a TOR implementation in which “bursts of 3 beeps are played 3 times, with silence in between burst sets”. In both examples the authors did not specify the duration of the pause between the beeps. In terms of duration (i.e., burst duration, pulse duration and total signal duration), the deducibility ranged between 10.0 % and 77.9 % ($M = 37.9\%$, $SD = 18.1$). Notably, the deducibility was below 52 % for visual and auditory classes, and above 60 % for the tactile vibration class. Finally, takeover time could be deduced in 56.3 % to 84.8 % across the classes ($M = 69.4\%$, $SD = 11.5$).

Location has also been coded in all classes. For visual and tactile classes the location was generally reported (deducibility between 71.9 % and 98.5 %). In contrast, studies did not often report where auditory signals were presented (19.6 % for speech; 35.4 % for earcons). Regarding the implementation of TORs, the types of imagery and sound used (or phrases in case of text and speech) were generally well-reported ($M = 89.0\%$, $SD = 3.9$). For visual classes the signal color(s) and the color of the surface on which the signal was displayed could on average be deduced in 79.5 % of the TORs ($SD = 13.1$). However, height, width, and distance were typically unreported, in which case the signal's angular size cannot be calculated. For text specifically, font type and font style were not reported in any of the studies, and could not be deduced based on photographs provided in the studies (which typically did provide enough detail to identify whether a serif font was used).

For auditory speech and earcons, as well as tactile vibration the signal intensity and its development over time were mostly unreported ($M = 6.3\%$, $SD = 7.1$), as a result of which the signal-to-noise ratio cannot be determined. Furthermore, earcon and vibration signals were not described in terms of onset and offset duration. The frequencies at which they were heard or felt were frequently omitted in the reporting (deducible in 42.4 % and 44.1 %, respectively). For speech signals, gender was reported in 36.6 % of the TORs, but intonation was not mentioned at all. Only one study ([Petermeijer et al., 2017b](#)) mentioned the size of the vibration sensors and their interspacing, based on which the vibration size could be calculated.

3.5.2. Implementation

Originally, a joint total of 88 unique phrases were found within the classes visual text and auditory speech, many of which are highly similar (e.g., ‘Take over’, ‘Take over!’, ‘Takeover’, and the German counterpart ‘Übernehmen’). Phrases with a similar meaning were clustered to reduce the number of categories. A phrase typically consisted of one or more elements of the following building blocks: [alert][reason][please][take over][control][immediacy][action]. For example, the phrase “Danger! Collision imminent you have control!” (Politis et al., 2017) was categorized as [alert][reason][take over][control] and the phrase “Alert! Take over vehicle control now!” (Roche et al., 2018) was categorized as [alert][take over][control][immediacy]. Alternatively to the above building blocks, phrases could concern a countdown (e.g., Wintersberger et al., 2018) or simply an action (e.g., “Press X” in Brandenburg & Chuang, 2019).

The color of a visual TOR and the pulse frequency of auditory and tactile TORs could be coded with up to three values. Signal color and signal color count were respectively calculated as all unique color combination and the number of colors used simultaneously. Signal frequency and signal frequency count were respectively calculated as all unique frequency combinations and the number of distinct frequencies used across pulses within a TOR. No differentiation was made between upward and downward frequency shifts (e.g., melodies in case of an earcon).

In the following, we focus on single attributes within each class, because the large number of unreported attributes impedes examination of combinations of attributes (as illustrated in Fig. 7 for classes). Only attributes of which a value is reported (deducible) in at least 50 % of the corresponding class are considered. Attributes of the visual text class are represented as circles in Fig. 10 (other classes are illustrated in Appendix E). The number of unique values in each attribute was calculated to obtain an initial overview of the degree of variability within each class. Certain attribute values have been used in very few studies (e.g., Pulse count: 7, $N = 1$), whereas other values have been re-used across several studies (e.g., Pulse count: 1, $N = 79$). Thus, the number of unique values alone is not enough to understand how the potential design space has been covered. Indicative for the distribution of unique values, the vertical axis in Fig. 10 shows the value with the largest proportion.

An attribute with at least two unique values can be said to have a dominant value if the value with the largest proportion covers >50 %. Marked by a dashed red line in Fig. 10, we distinguish between attributes with low variability (=less than 3 unique values, or a dominant value) and attributes with high variability (>3 unique values and no dominant value). Table 2 presents an overview of the variability of attributes across classes after applying the same criterion to the other classes (see Appendix E). Such information may be helpful in future meta-studies when considering certain attributes as factors. For example, phrase may be difficult to operationalize as dichotomous variable, whereas language may feature too little variability to perform inferential statistics.

Four attributes only showed a low variability. Across all classes, a burst count of 1 has been reported exclusively (text, symbol, abstract, speech), or almost exclusively (earcon: 98.1 %, vibration: 95.9 %). Consequently, burst interval only applied to one earcon ($t = 14000$ ms) and three TORs with vibration (twice at $t = 800$ ms and once at $t = 1800$ ms). Phrases were predominantly reported in English for speech (96.6 %) and text (82.9 %). It should be noted that phrases were occasionally reported in English, even when non-English participants were recruited. This raises the question to which extent the actual phrase language (e.g., German) was biased by the language in which the manuscript was published (i.e., English). Within TORs with a text class, 98.8 % used a sans serif font.

Seven attributes only showed a high variability. Takeover time met this criterion in all classes and showed between 11 and 26 unique values (see Appendix E). The average takeover time per class ranged between $M = 7000$ ms for earcons ($SD = 4970$) and $M = 11470$ ms for vibration ($SD = 16560$). Regarding signal color, TORs with texts were often, but not dominantly, red (34.0 %) or white (35.9 %), and likewise, TORs with abstract visuals were often red (42.2 %). For symbolic TORs unicolor red was the most common implementation (24.0 %), followed by a combination of black and white (16.7 %). The most common pulse durations in abstract TORs were 100 ms and 250 ms ($M = 5045$ ms, $SD = 9756$). For TORs with tactile vibration, pulse durations of 200 ms were often found ($M = 254$ ms, $SD = 184$), but not enough to consider it a dominant implementation. Burst duration ($M = 3735$ ms, $SD = 5662$) and total duration ($M = 8039$ ms, $SD = 10188$) also showed high variability in vibration TORs, but did not meet the criterion for deducibility in other classes. Phrase showed a high variability: the four (text) and five (speech) largest phrase categories constituted the first 50 % of the reported phrases. These phrase categories were: [take over], [please][take over], [take over][control], [alert][reason] (speech only), and [alert][reason][take over][control]. As a result, the number of syllables in speech also showed a high variability. Ranging between 1 and 22 syllables, the most occurring value was 5 syllables (14.4 %), followed by 12 syllables (13.3 %).

Finally, six attributes showed high variability in some classes and low variability in other classes. Starting with pulse count, a single pulse is mostly found in symbols (93.2 %), speech (91.4 %), and text (82.2 %), occasionally alternated with 2 or 3 pulses (symbols and speech) or a larger series of pulses (text) associated with countdown timers (e.g., Wu et al., 2019b). A relatively large variety in pulse count is found for classes abstract (1: 56.4 %, 8: 25.6 %), earcon (1: 46.2 %, 2: 27.4 %, 8: 8.5 %) and vibration (1: 28.9 %, 3: 31.1 %, 8: 17.8 %). All TORs with 8 pulses were implemented by Politis et al. (2017), indicating that a single study played a relatively large role in the variability of pulse count. The pulse interval was exclusively 0 ms for text and symbol TORs and dominantly 100 ms for abstract (62.5 %) and earcon (63.6 %) TORs. In vibration TORs 100 ms was the most occurring value, but it was not dominant (38.5 %, $M = 180$ ms, $SD = 161$). In speech TORs, no single pulse interval value occurred more than once ($M = 6800$ ms, $SD = 2864$). Regarding location, on the one hand TORs with vibration were situated mainly in the chair (71.6 %), followed at a distance by wristbands (26.9 %) and the steering wheel (1.5 %). On the other hand, visual TORs showed a larger variability, where at least two (symbols) or three (text, abstract) categories were required to make up for 50 % of the reported locations (typically involving the center console, windshield, instrument cluster, Head Up Display (HUD), or a combination of these). In terms of material, dominant categories were

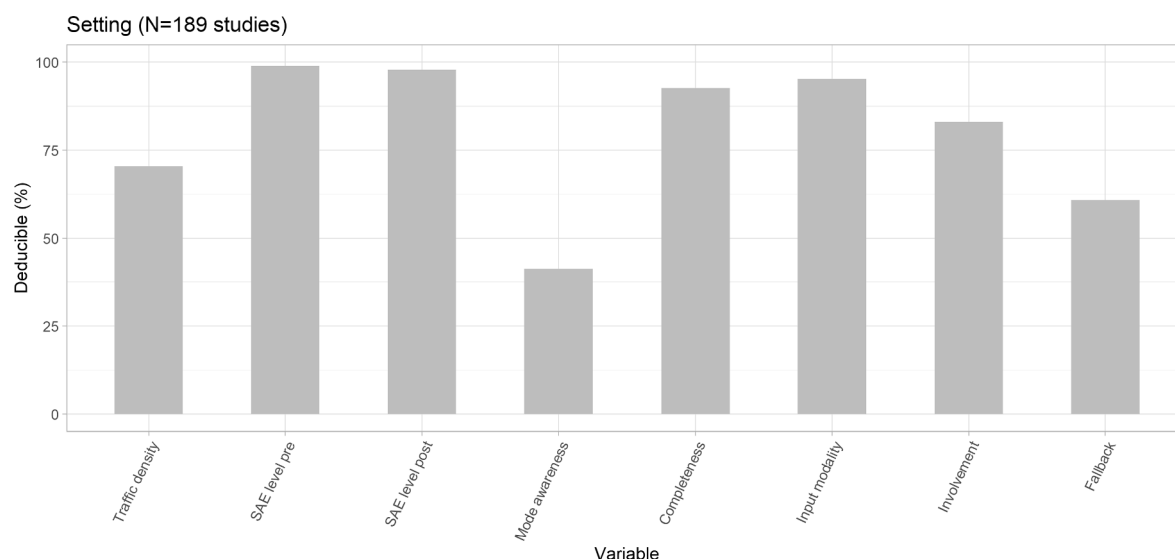


Fig. 12. Deducibility of setting variables. Note: setting variables do not depend on TOR implementation and therefore apply to all studies.

found at symbols, which mostly involved hands placed on a steering wheel (59.6 %) followed by a combination of symbols (17.3 %, e. g., hands on wheel and foot on pedal), and at earcons, typically consisting of unspecified ‘beeps’ or ‘tones’ (75.4 %) followed by sinusoids (18.1 %). TORs with an abstract class, on the other hand, did not feature a dominant category: line (25.0 %), circle (20.0 %), rectangular (11.7 %), point (11.7 %), other (31.7 %). Regarding the number of signal colors, text (100 %) and abstract (71.9%) TORs typically featured one signal color (71.9 %) (i.e., low variability). Unicolor, bicolor and tricolor showed similar proportions in symbolic TORs (high variability). TORs with an abstract class mostly featured a transparent background color (67.4 %), whereas text and symbols involved eight different colors; black (31.5 % and 35.6 %, respectively), red (26.1 % and 15.1 %) and transparent (25.0 % and 19.2 %) being the largest categories.

3.6. Background signals

Background signals were coded in terms of NDRT-involvement (for each study) and in terms of non-NDRT-related signals (separately for each modality if used within a TOR). The deducibility of background signals is presented in Fig. 11, with an average of $M = 43.7$ % ($SD = 40.9$). In general, the presence of NDRT-involvement could be well deduced, although auditory NDRT-involvement proved to be more difficult to derive than visual and tactile NDRT-involvement. For example, when participants had to play a game on a tablet, it was often unclear if the game used audio and whether the tablet emitted those sounds. If deducible, visual, auditory, and/or manual (tactile) NDRT-tasks were performed in at least one experimental condition in 85.2 %, 38.1 %, and 72.6 % of the studies, respectively.

Presence of visual background signals (aside from NDRT-related signals) could be deduced in 36 studies featuring a TOR with a visual component (27.3 %), within which 26 studies with at least one visual background signal was identified. For example, in Naujoks et al. (2015) visual-auditory TORs related to a traffic jam assist were accompanied by visual cues indicating the status of several subcomponents of the traffic jam assist. These signals were subsequently coded on whether a background signal was related to vehicle automation (as was the case in the example of Naujoks et al., 2015), as well as Spatial and Temporal proximity in relation to the TOR location and timing. A relation with vehicle automation was deducible in 24 out of 26 visual background signals, and true for 20 signals. Spatial proximity was deducible in 6 signals, always showing a distance of 0 cm. Here it should be noted that distances other than 0 cm could not be derived based on photographs of the experimental setup, resulting in an ‘unknown’ value. Temporal proximity was deducible in 19 signals, always coinciding with the TOR.

The auditory background was coded in terms of noise (e.g., engine, road, wind) and in terms of non-NDRT signals (e.g., automation status messages). Occasionally studies reported which sound system was used as part of a driving simulator setup (e.g., Bourelly et al., 2019). Seeing that a driving simulator can be configured by the experimenter, including which sounds are presented to a participant, auditory background signals were only coded when explicitly mentioned. Consequently, noise was deducible and present in 44 studies featuring a TOR with an auditory component (26.3 %). Noise intensity was reported in 2 of those 44 studies, namely engine noise of 60 dB (van den Breukel et al., 2016) and road noise of 50 dB (Mok et al., 2017). In 9 studies where a TOR included an auditory component it was possible to deduce whether an auditory (non-NDRT, non-noise) background signal was present (5.4 %), which was the case in 5 studies. For as far as could be deduced, in 2 studies the corresponding signal(s) had a relation with automation, and in two other studies it did not. Presentation location and timing, and hence spatial and temporal proximity, could not be deduced.

None of the 28 studies reporting a TOR with a tactile component provided information about the intensity of tactile background noise.

Detectability and discriminability of a TOR signal cannot be assessed when studies do not report the presence and intensity of background signals, or fail to report the intensity of a TOR signal (see above). This lack of information was found in the majority of the studies. It should be noted however, that a small portion of the studies did report about the detectability of a TOR without explicitly stating the signal and background intensity. For example, [Li et al. \(2020\)](#) did not report the loudness of an auditory TOR and its background, but did report the TOR was loud enough to be perceived over entertainment video sounds. The coding sheets could not account for such descriptions in the present study, resulting in missing values. Adjustment of the coding scheme may be required for a future inventory.

Another observation is that studies reporting (background) signal intensity mostly expressed intensity in the acoustical unit dB, but [Politis et al. \(2015\)](#) used the electrical unit dBFS. The latter unit does not provide an indication on the intensity with which participants heard the TORs. Furthermore, studies expressing intensity in dB did not report where such measurements were taken (e.g., at the listening position), nor how such measurements were obtained (with notable exceptions found in [Petermeijer et al., 2016](#) and [Petermeijer et al., 2017c](#)). Consequently, the validity of a comparison of TOR intensities across studies is unknown.

3.7. Setting

Deducibility of the setting variables fluctuated between 41.3 % and 98.9 % ($M = 80.0$ %, $SD = 20.8$), see [Fig. 12](#). TORs were often ($N = 107$, corresponding with 80.5 % of the deducible sample) issued in presence of other traffic in the neighboring lane(s). Traffic density, expressed as the number of vehicles per kilometer (following the meta-analysis of [Gold et al., 2018](#)), could be derived in 20 studies. Several studies used alternative measures, such as a qualitative indication (e.g., ‘moderate traffic conditions’ in [Eriksson et al., 2017c](#)) or the number of vehicles per hour (e.g., [Lin et al., 2020](#); [Wu et al., 2020](#); [Vogelpohl et al., 2019](#); [Blommer et al., 2017](#)). Such studies were coded as ‘traffic present, number unspecified’. In terms of SAE levels, the majority of the studies ($N = 173$, corresponding with 93.5 % of the deducible SAE levels) described a transition from level 3 (conditional) or level 4 (high) to level 1 (assisted) or level 0 (manual). The lowest deducibility within setting was found for mode awareness (41.3 %), which is alarming given the importance of clear communication (e.g., [Carsten & Martens, 2019](#)). In so far as communication on the automation status prior to the TOR was reported ($N = 78$), most of the studies used at least a symbol ($N = 50$; 64.1 % of the deducible sample), especially in exclusive combination with color ($N = 25$; 32.1 %), such as a car icon with a green road ahead in [Lu et al. \(2019\)](#). Regarding completeness of transfer, all driving functions were typically transferred at the same time ($N = 165$; 94.3 %). Drivers almost exclusively responded to transition requests through physical controls (e.g., steering wheel, pedals, buttons, levers; $N = 174$; 96.7 %). An exception was a dialogue based TOR described by [Politis et al. \(2018\)](#), aiming to increase situation awareness prior to a non-critical takeover situation. Furthermore, four studies involved a manipulation of the input modality. For example, [Wintersberger et al. \(2019\)](#) describe two experiments; one in which participants had to respond to a TOR by putting their hands on the steering wheel, and another in which a touch tablet was used as substitute for the steering wheel. In most cases, drivers did not have to confirm that they took over control (i. e., involvement; $N = 139$; 88.5 %). If drivers did not respond, mostly no fallback was implemented, meaning that a potential crash was not automatically avoided ($N = 87$; 75.7 %). A total of 26 studies reported that the vehicle automatically stopped (in own lane $N = 15$; outside own lane $N = 2$; lane unknown $N = 9$).

4. General discussion

Meta studies have largely neglected the influence of TOR design on take-over performance. The aim of this study was to inform future meta studies about the level of detail with which TOR designs are reported, by systematically reviewing academic literature with a newly postulated categorization framework. The main findings, implications, and limitations of the systematic review, along with recommendations for future research, will be presented first. Next, improvements and extensions of the framework are discussed.

4.1. Systematic review

4.1.1. Main findings & implications

Our systematic review classified a total of 391 TOR designs found in 189 academic studies. Compared to previous inventories of TOR designs (35 TORs in [Mirnig et al., 2017](#); 31 TORs in [Boelhouwer et al., 2019](#)), the present inventory is the largest in terms of sample size as well as the number of coded variables. In general, deducibility of variables decreased when variables resided deeper within the hierarchical framework presented in [Fig. 1](#). This decreasing order was also found in the percentage agreement between coders, suggesting that a lack of reporting fosters ambiguity when interpreting an experimental setup.

To analyze the effect of TOR design on take-over performance, it is imperative that studies investigating take-over situations describe the TOR design at a detailed level. Prospective meta studies may benefit from the finding that deducibility was high at the levels of Modality (i.e., 98.7 %, included in previous meta studies) and Class (87.1 % on average, not included so far). The same cannot be said for attributes, however. With an average attribute deducibility of 50.5 % across all deducible classes, the sample that can be included in a meta study involving TOR design at the level of attributes may be too low to reach a desirable statistical power. There are more implications of a low deducibility of TOR designs. For example, researchers intending to replicate a study may not be able to

reproduce the exact same experimental setup. Lacking insights from meta studies, the impact of deviations from a former experimental setup may remain unknown. As another example, researchers may not be able to assess compliance with HMI guidelines, such as the use of contrasting colors and minimum signal-to-noise ratios to ensure detectability (e.g., [Naujoks et al., 2019b](#)).

In terms of implementation, no archetypical TOR appears to exist. Visual-auditory TORs may constitute the largest group in terms of modality, but at class level there is no clearly favorable TOR design. Auditory TORs, the second-largest group in terms of modality, features either earcons or speech. Other than noting that these classes are single bursts (earcons and speech), involving beeps or unspecified tones (earcons), or English phrases (speech), no further common themes can be derived from the multitude of coded variables. The reasons for a lack of common themes are low deducibility at the attribute level, and a large variation within attributes in case deducibility is high.

Another finding in terms of implementation is that a large part of the potential TOR design space has not been covered (or coverage is unknown due to a low deducibility). For example, almost all TORs involved a single burst, which suggests that temporal or rhythmic aspects of TORs have received little attention. Also, the classes sonification, anthropomorphism, and force feedback were not encountered, and only a single study reported auditory icons. Knowing which classes constitute a sample helps to interpret the findings of previous meta studies. For example, [Zhang et al. \(2019\)](#) report that unimodal visual TORs were associated with longer mean TOTs than unimodal auditory or vibrotactile TORs. This finding may be true for text, symbols, and abstract unimodal visual TORs, albeit potentially biased by text-based TORs (see [Fig. 9](#)). This finding may also be true for earcons, speech, and to a lesser extent for auditory icons (due to its low prevalence). However, the studies examined in [Zhang et al. \(2019\)](#) did not include TORs with sonification, nor anthropomorphism, and effects of these classes on TOT therefore remain unknown. Thus, one should be cautious making statements that generalize towards an entire modality, instead of the classes that constitute the sample.

4.1.2. Limitations & recommendations

The observation that certain classes have not been implemented (or described) and that certain attributes show little variation does not necessarily mean that implementing those classes and/or choosing other implementations on certain attributes will improve take-over performance. Such impact is largely unknown as of yet. Thus, follow-up research is recommended in which, amongst others, TOR class is systematically manipulated, involving classes that have received no attention (i.e., anthropomorphism, sonification, force feedback) or little attention (e.g., auditory icons).

There are seven factors that have or may have contributed to a lowered deducibility. First, the hierarchical framework poses restrictions in that if a TOR attribute is known, but its class is not, the attribute cannot be coded (e.g., pulse duration could belong to any auditory class). Second, attributes were at times reported using different units than what was accounted for in the coding sheet (e.g., traffic density in vehicles per hour versus vehicles per kilometer). Third, attributes (e.g., signal and background loudness) may have been unknown, but the goal of reporting these attribute (e.g., detectability) was met nonetheless. Fourth, several studies referenced to previous work when reporting a TOR implementation. In case of ambiguous references (using phrases such as ‘based on...’ without further specifications) it is unknown if and which deviations from the previous setup have been introduced. Such ambiguity should therefore be avoided when reporting an experimental setup. Seven studies unambiguously referred to previous work. Modality and class could typically be derived without consulting the referenced materials, but the possibility exists that further information may be found on attributes, setting, and/or background related variables, which means deducibility figures reported in our study may be an underestimation. Fifth, article length restrictions may have forced authors to leave out certain details about the experimental setup. Sixth, studies involving an experimental manipulation of the TOR design more likely report details on (differences between) the TOR designs than studies that do not involve such a manipulation. Seeing that many studies in our sample did not involve an experimental manipulation of the TOR design, a bias towards a lowered deducibility may have been introduced. Seventh, and related to the former factor, authors may have been unaware about the relevance of certain details for future studies when writing the manuscript ([National Academies of Sciences, Engineering, and Medicine, 2019](#)). Factors 1–4 may be mitigated by adjustments in the coding scheme or framework, or by examining referenced studies. Factors 5–7, however, require more awareness about the need for detailed documentation, e.g., by optimizing guidelines for reporting an experimental setup.

A bias may have been introduced by our information sources, search strategy, eligibility criteria, and study selection method. More studies may have been found when other databases (e.g., IEEE, Scopus, Google Scholar) were consulted in addition to the information sources consulted in the present study, including more theses, books, conference papers and unpublished materials ([Martín-Martín et al., 2018](#)). Nonetheless, studies examined in the meta-analyses of [Zhang et al. \(2019\)](#) and [Weaver and DeLucia \(2022\)](#), as well as the reviews of [Mirnig et al. \(2017\)](#) and [Boelhouwer et al. \(2019\)](#) were collected with similar queries as the present study, using Google Scholar, ResearchGate, PsycINFO, PsycArticles, MEDLINE, ACM Digital Library, IEEE, ScienceDirect, Scopus, as well as using snowballing techniques. Thus, indirectly, the present study made (partial) use of those information sources. We therefore expect that our sample is representative for TOR designs published in English academic literature. However, it is not complete. The meta-analysis and framework studies, each of which involved queries using several databases, were published before 2021. This means that our sample may exclude studies published in 2021 if unavailable through Web of Science. Also, studies published in languages other than English were not included. Additional databases and languages need to be included to obtain a complete sample.

Another bias may have been introduced by the possibility that multiple reports of the same study were included, such as a conference proceeding re-submitted to a journal using a different title. [Dey et al. \(2020\)](#) note that information about a single design is sometimes described over multiple reports. Our rationale is that if one reads a publication, one should not have to search for other

potentially existing reports of the same study to get a full picture of the experimental design, unless an explicit reference to previous work is provided. Several reports referred to previous work when describing a TOR design. However, in case of ambiguous references in which the TOR design was ‘based on’ previous work, one cannot exclude the possibility that subtle differences in the experimental setup have been introduced. Judging whether such differences are large enough to speak of distinct TOR designs is prone to interpretation (Dey et al., 2020). From the perspective of deducibility, and to minimize bias introduced by interpretation, we chose to include all reports, even if they may have described the same study. By doing so, descriptive statistics on the implementation of a TOR design may have suffered from a bias due to double counting. Future meta studies need to account for such bias by applying more strict study selection criteria.

4.2. Reflection on the framework

4.2.1. Main findings & implications

The framework introduced in the present study extends previous HMI classification frameworks (Mirnig et al., 2017; Boelhouwer et al., 2019; Dey et al., 2020) and takes into account parameters required to assess HMI guidelines reported by Naujoks et al. (2019b), by introducing a set of unique classes within each modality and a set of generic attributes (e.g., location, intensity, frequency, duration) as well as class-specific attributes. Consequently, the present framework involves more variables at a finer level of detail than previous classification frameworks, allowing for detailed inventories of and comparisons between TOR designs. The increased level of detail also allows for more accurate replication of experimental setups in future studies.

A pre-requisite for inventorying and comparing TOR designs (e.g., as part of a meta-analysis on take-over performance), as well as replicating an experimental setup, is that studies actually report the details described in the framework. The notion of missing data (i.e., lowered deducibility) is not new (e.g., Joscelyn & Maickel, 1977; Cabral & Remijn, 2019). Previous inventories of TOR designs also included missing values (Mirnig et al., 2017), or used ‘unspecified’ as one of the coding options (Dey et al., 2020). Medical research typically makes use of checklists to facilitate full reporting of experimental stimuli for future replication (e.g., Sakzewski et al., 2016; Avey et al., 2016). Research on traffic safety may similarly benefit from such a checklist, which at present appears to be missing. The framework postulated in the present study and the accompanying codebook may assist in systematically reporting TOR design parameters in future studies (e.g., by including a filled out checklist as an appendix), and to create an inventory of TOR design parameters used in previous studies.

4.2.2. Limitations & recommendations

The systematic review presented here is a first exploration of the classification framework. The low interrater reliability on a number of variables indicates that clearer operational definitions may be needed. What should be realized, though, is that information on an experimental setup was gathered by researchers not involved in those studies, based on published reports only. Moreover, interrater reliability was lower at deeper levels of the framework, where information was more often found to be lacking. Interpretation of the operational definitions may be less ambiguous if studies are coded by researchers at the time they are involved in the corresponding studies, with access to (additional documents about) the experimental setup. Therefore, a follow-up study is recommended to better assess the interpretability of the operational definitions, by having researchers code their own studies.

The framework can be extended in terms of depth by including the possibility to categorize modulations of certain TOR attributes while TORs are presented, seeing that such modulations may influence take-over performance. For example, urgency can be conveyed by decreasing the inter-burst interval or inter-pulse interval (Van der Heiden et al., 2017; Van den Beukel et al., 2016), an approach also used by parking sensors to communicate gap size. Tuuri et al. (2011) show that urgency can also be conveyed by applying intonation to earcons (e.g., changing the pitch of a tone during a pulse). Furthermore, Imbert et al. (2014) show that animation and movement of visual notifications influence detection speed and accuracy by air traffic controllers. Possibly these effects are also true for TOR signals in an automated driving context. Additionally, certain TOR attributes may be designed to adapt to environmental conditions (e.g., a louder TOR in a noisy driving environment), and vice versa, certain environmental conditions may adapt to the presence of a TOR (e.g., pausing music prior to a TOR). Further extensions in terms of depth may include the number of TOR exposures during an experiment and whether TORs have been manipulated in terms of modality and/or class, to accommodate studies on learning effects, habituation, and potential communication of TOR reasons.

The framework can also be extended in terms of scope. The present framework and systematic review focused on the context of a takeover scenario. Other use cases involving HMI signals may be classified with relatively minor adaptations of the framework. For example, Mulvihill et al. (2021) describe four implementations of a driver state monitoring and warning systems. One of those implementations involves an ‘auditory alerting tone’ at 70 dBA, followed by a spoken message with an Australian female voice at 70 dBA and a tactile warning with 5 pulses. In our framework, this implementation would be classified as auditory-tactile (modality), with an earcon, speech, and vibration (classes). The intensity of the auditory components as well as the phrase can be deduced, but earcon frequency and duration cannot (attributes). Furthermore, the traffic density of the driving simulator is mentioned (setting), as well as the fact that participants were distracted by a visual-manual NDRT (background signals). Thus, the hierarchical structure introduced in Fig. 1 may be re-used in other use cases, except that the labels ‘Takeover’ and ‘TOR signal’ should be replaced by the corresponding use case and main signal names, respectively. Additional research is needed to determine to what degree individual variables within each

block of the hierarchical structure apply, and if additional levels within categorical variables are required in alternative use cases.

When considering concurrent use cases (e.g., a takeover scenario in combination with driver state monitoring and warning system), simultaneous presentation of signals originating from separate sub-systems may become an issue for discriminability, detectability, and understandability. For example, two sounds with similar acoustic properties may be perceived as a single auditory object (Bregman, 1994), and accuracy in identifying individual sounds decreases when more sounds are presented concurrently (Nees & Walker, 2011). Naujoks et al. (2019b) present several guidelines related to discriminability and detectability of visual, auditory, and tactile HMI signals, including advisory ranges for pulse duration and frequency, as well as raising the intensity of auditory signals above noise produced by a vehicle in motion. However, no guidelines are presented on avoiding (perceptual) confusion between multiple signals. In the present framework, detectability and discriminability of a single signal can be examined relative to all background signals. In the takeover use case, signals related to driver monitoring and warning systems may be part of the background. Vice versa, in the use case of driver monitoring and warning, TOR signals may be part of the background. Thus, our framework may assist the (design or) evaluation of an environment in which multiple signals are presented concurrently.

Finally, the framework could be extended to facilitate principles of inclusive design. For example, restrictions in sight, hearing and touch may limit the detectability of a TOR, which otherwise would have been perfectly detectable for drivers without such restrictions. Furthermore, cultural differences have been shown to influence, e.g., meaning attributed to colors (Tham et al., 2020) and HMI usability (Khan & Williams, 2014). Adding participant characteristics to the framework (e.g., as part of the ‘Setting’) could facilitate future meta-analysis to investigate potential moderating effects of the population on the relation between TOR design and takeover performance, and subsequent design guidelines that may result from such meta-analysis.

5. Conclusion

Meta studies on factors contributing to take-over performance did not systematically include TOR design, other than the modality at which TORs are presented. A detailed understanding of the influence of TOR design on take-over performance is therefore lacking. The present study postulated a classification framework to facilitate a detailed description of TOR designs. The framework was operationalized into a coding sheet to perform a systematic review on TOR designs in academic literature. The systematic review revealed that no predominant TOR design exists, and a considerable part of the available design space has not yet been explored. Moreover, the deducibility of reported information on TOR designs appears to be inversely related with the level of detail at which TORs are examined. The classification framework and the corresponding coding sheet could support systematic reporting in future work. If the devil is indeed in the details, subsequent meta-analysis will provide a better understanding about the impact of TOR design on take-over performance, which in turn can support implementation of safe and effective TORs in our vehicles.

CRediT authorship contribution statement

Reinier J. Jansen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Supervision. **Angelica M. Tinga:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Rins de Zwart:** Methodology, Investigation, Writing – review & editing. **Sander T. van der Kint:** Methodology, Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Coded studies are provided as [supplementary materials](#).

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Appendix A.: PRISMA checklist

See [Table A1](#).

Table A1

Checklist of the preferred reporting items for systematic reviews and meta-analyses (PRISMA).

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section 1
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section 1.2
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 2.2
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 2.1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 2.1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2.3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 2.3
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 2.4.1, Appendix B
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 2.4.1, Appendix B
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2.4.2
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Section 2.4.1
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Section 2.3
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 2.4.1
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 2.4.1
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 2.4.1
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, <i>meta</i> -regression).	Not applicable
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not applicable
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Section 2.1
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Section 2.3
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 2.3
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Section 3.1
Study characteristics	17	Cite each included study and present its characteristics.	Appendix C
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section 3.1 , Section 4.1.2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Section 3 , Appendix C
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section 3 , Section 4.1.2

(continued on next page)

Table A1 (continued)

Section and Topic	Item #	Checklist item	Location where item is reported
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Section 3
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not applicable
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not applicable
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Section 4.1.2
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Section 3.5.2 , Section 4.1.2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section 4.1.1
	23b	Discuss any limitations of the evidence included in the review.	Section 4.1.2
	23c	Discuss any limitations of the review processes used.	Section 4.1.2 , Section 4.2.2
	23d	Discuss implications of the results for practice, policy, and future research.	Section 4
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Section 2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Section 2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not applicable
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Acknowledgements
Competing interests	26	Declare any competing interests of review authors.	Acknowledgements
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Appendix B,S upplementary materials

Appendix B:. Codebook

See [Table B1](#).

Table B1

Codebook of the classification framework. Note: unreported frequency data or continuous data has been coded with value ‘-1’ and has been interpreted as ‘not deducible’ in subsequent analysis.

Variable	Explanation variable	Categories	Explanation categories
Modality			
TOR Visual	At least one TOR was VISUAL only.	Yes, No, Unknown	
TOR Auditory	At least one TOR was AUDITORY only.	Yes, No, Unknown	
TOR Tactile	At least one TOR was TACTILE only.	Yes, No, Unknown	
TOR bimodal: Visual + Auditory	At least one TOR was bimodal with a VISUAL and an AUDITORY component.	Yes, No, Unknown	
TOR bimodal: Visual + Tactile	At least one TOR was bimodal with a VISUAL and a TACTILE component.	Yes, No, Unknown	
TOR bimodal: Auditory + Tactile	At least one TOR was bimodal with an AUDITORY and a TACTILE component.	Yes, No, Unknown	
TOR trimodal	At least one TOR was trimodal with a VISUAL, and AUDITORY, and a TACTILE component.	Yes, No, Unknown	
Class			
Visual text	Does the visual TOR have a text component? Explicit use of text, like “Take over!”, or any other variation that uses text to communicate vehicle state or instructions to the driver.	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Visual anthropomorphic	Does the visual TOR have an anthropomorphic component? The concept uses human-like or anthropomorphic elements for communication, such as facial expressions (eyes, smiling), or gestures.	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Visual symbolic	Does the visual TOR have a symbolic component? Use of recognizable (traffic) symbols such as stop signs, arrows, hands on a steering wheel, or other forms of symbols used to communicate. (note: we do not adopt the semiotic interpretation of symbols, icons and indices)	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Visual abstract	Does the visual TOR have an abstract component? Abstract visual shapes or other forms of light-based or non-light-based communication devices that use metaphors, or signals that aim at communicating intuitively via a nonconcrete, open-to-interpretation interface that is neither anthropomorphic, textual, or symbolic.	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Auditory speech	Does the auditory TOR have a speech component? Prerecorded natural voicings or synthetic voicings.	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Auditory icon	Does the auditory TOR have an auditory icon component? Brief sounds that have some a priori association with their referent object, event, or process. The relationship between the sound and its meaning in the interface can range from quite literal (e.g., the sound of crackling flames to represent an engine fire) to more metaphorical (e.g., the sound of crumpling paper to represent a computer file being deleted).	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Auditory earcon	Does the auditory TOR have an earcon component? Earcons are abstract, artificial sounds that bear no ecological relationship to the represented process or event (e.g., a single ‘beep’ or ‘tone’).	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Auditory sonification	Does the auditory TOR have a sonification component? The use of non-speech audio to display quantitative data (e.g., geiger counters, parking assist systems).	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Tactile vibration	Does the tactile TOR have a vibration component? (e.g., vibration implemented in the seat or steering wheel)	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.
Tactile force feedback	Does the tactile TOR have a force feedback component? (e.g., resistance applied to the pedals)	Yes, No, Unknown, Not applicable	A class is not applicable if the corresponding modality is not implemented.

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
Attributes (visual text)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Location	Where is the signal presented?	Instrument cluster Dashboard Center console HUD Windshield Handheld device Steering wheel Door (s) Combination Other Unknown	
Language	The language of the phrase.	English German Chinese French Spanish Dutch Other Unknown	
Phrase	The phrase as presented to the driver.	N.A.	
Background color	The color immediately surrounding the text.	Black White Grey Yellow Orange Red Purple Green Cyan Blue Violet Transparent Other Unknown	
Text color	The color of the text	Black White Grey Yellow Orange Red Purple Green Cyan Blue Violet	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
		Transparent	
		Other	
		Unknown	
Phrase width	Width of the phrase as presented on the visual display	Unit: mm	
Phrase height	Height of the phrase as presented on the visual display	Unit: mm	
Distance to driver	Distance between signal center and driver's eyes	Unit: cm	
Text serif	Is the text using a serif typeface?	Yes, No, Unknown	
Font type	Font type	E.g., Arial, Times	
		New Roman	
Font style	Font style	Regular	Select 'Unknown' of font type is unknown
		Italic	
		Bold	
		Bold italic	
		Light	
		Light italic	
		Condensed	
		Condensed italic	
		Condensed bold	
		Other	
		Unknown	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding a blank display between bursts)	Unit: ms	
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Phrase count	How often is the phrase repeated to the driver? (a.k.a., pulse count)	Integer	
Phrase duration	Duration of a single phrase. (a.k.a., pulse duration)	Unit: ms	
Inter-phrase interval	Time (silence) between phrases (in case of phrase count > 1; a.k.a., inter-pulse interval)	Unit: ms	
Attributes (visual anthropomorphic)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Location	Where is the signal presented?	Instrument cluster	
		Dashboard	
		Center console	
		HUD	
		Windshield	
		Handheld device	
		Steering wheel	
		Door	
		(s)	
		Combination	
		Other	
		Unknown	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
Material	What kind of imagery is used to make it human-like?	Face Hands Face & Hands Other Unknown	
Background color	The color immediately surrounding the signal	Black White Grey Yellow Orange Red Purple Green Cyan Blue Violet Transparent Other Unknown	
Signal color (main)	The main color of the signal	Black	Category 'Not applicable' can be used for secondary and tertiary colors in case only one (or two) signal colors are used. The order of main/secondary/tertiary is arbitrary.
Signal color (secondary)	The secondary color of the signal (in case of multiple colors)	White	
Signal color (tertiary)	The tertiary color of the signal (in case of multiple colors)	Grey	
		Yellow	
		Orange	
		Red	
		Purple	
		Green	
		Cyan	
		Blue	
		Violet	
		Transparent	
		Other	
		Unknown	
		Not applicable	
Signal width	Width of the signal as presented on the visual display	Unit: mm	
Signal height	Height of the signal as presented on the visual display	Unit: mm	
Distance to driver	Distance between signal center and driver's eyes	Unit: cm	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding a blank display between bursts)	Unit: ms	
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Pulse count	How often is the signal repeated to the driver?	Integer	
Pulse duration	Duration of a single pulse (excluding non-display time)	Unit: ms	
Inter-pulse interval	Time between pulses (in case of pulse count > 1)	Unit: ms	
Attributes (visual symbol)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Location	Where is the signal presented?	Instrument cluster Dashboard Center console	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
		HUD	
		Windshield	
		Handheld device	
		Steering wheel	
		Door	
		(s)	
		Combination	
		Other	
		Unknown	
Material	What kind of shape is used?	Hands on wheel	
		Traffic sign	
		Foot on pedal	
		Arrow	
		Combination	
		Other	
		Unknown	
Background color	The color immediately surrounding the signal	Black	
		White	
		Grey	
		Yellow	
		Orange	
		Red	
		Purple	
		Green	
		Cyan	
		Blue	
		Violet	
		Transparent	
		Other	
		Unknown	
Signal color (main)	The main color of the signal	Black	Category 'Not applicable' can be used for secondary and tertiary colors in case only one (or two) signal colors are used. The order of main/secondary/tertiary is arbitrary.
Signal color (secondary)	The secondary color of the signal (in case of multiple colors)	White	
		Grey	
Signal color (tertiary)	The tertiary color of the signal (in case of multiple colors)	Yellow	
		Orange	
		Red	
		Purple	
		Green	
		Cyan	
		Blue	
		Violet	
		Transparent	
		Other	
		Unknown	
		Not applicable	
Signal width	Width of the signal as presented on the visual display	Unit: mm	
Signal height	Height of the signal as presented on the visual display	Unit: mm	
Distance to driver	Distance between signal center and driver's eyes	Unit: cm	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding a blank display between bursts)	Unit: ms	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Pulse count	How often is the signal repeated to the driver?	Integer	
Pulse duration	Duration of a single pulse (excluding non-display time)	Unit: ms	
Inter-pulse interval	Time between pulses (in case of pulse count > 1)	Unit: ms	
Attributes (visual abstract)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Location	Where is the signal presented?	Instrument cluster Dashboard Center console HUD Windshield Handheld device Steering wheel Door (s) Combination Other Unknown	
Material	What kind of shape is used?	Circle Square Rectangular LinePoint (s) Other Unknown	
Background color	The color immediately surrounding the signal	Black White Grey Yellow Orange Red Purple Green Cyan Blue Violet Transparent Other Unknown	
Signal color (main)	The main color of the signal	Black	Category 'Not applicable' can be used for secondary and tertiary colors in case only one (or two) signal colors are used. The order of main/secondary/tertiary is arbitrary.
Signal color (secondary)	The secondary color of the signal (in case of multiple colors)	White	
Signal color (tertiary)	The tertiary color of the signal (in case of multiple colors)	Grey	
		Yellow	
		Orange	
		Red	
		Purple	
		Green	
		Cyan	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
		Blue	
		Violet	
		Transparent	
		Other	
		Unknown	
		Not applicable	
Signal width	Width of the signal as presented on the visual display	Unit: mm	
Signal height	Height of the signal as presented on the visual display	Unit: mm	
Distance to driver	Distance between signal center and driver's eyes	Unit: cm	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding a blank display between bursts)	Unit: ms	
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Pulse count	How often is the signal repeated to the driver?	Integer	
Pulse duration	Duration of a single pulse (excluding non-display time)	Unit: ms	
Inter-pulse interval	Time between pulses (in case of pulse count > 1)	Unit: ms	
Attributes (auditory speech)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Location	Where is the signal presented?	Instrument cluster	
		Dashboard	
		Center console	
		HUD	
		Windshield	
		Handheld device	
		Steering wheel	
		Door	
		(s)	
		Combination	
		Other	
		Unknown	
Phrase intensity	Intensity of the first phrase (excl. silence between phrases)	Unit: dB	
		dB(A)	
		dB(B)	
		dB(C)	
		dB SPL	
		dBFS	
		dBV	
		dBu / dBv	
		Knob settings	
		Other	
		Unknown	
Language	The language of the phrase.	English	
		German	
		Chinese	
		French	
		Spanish	
		Dutch	
		Other	
		Unknown	

(continued on next page)

Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
Gender	Gender of the human or synthetic voice.	Female Male Other Unknown	Use the ‘/’ symbol to indicate sequential messages.
Phrase	The phrase as presented to the driver.	Text	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding silence between bursts)	Unit: ms	
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Phrase count	How often is the phrase repeated to the driver? (a.k.a., pulse count)	Integer	
Phrase duration	Duration of a single phrase. (a.k.a., pulse duration)	Unit: ms	
Inter-phrase interval	Time (silence) between phrases. (a.k.a., inter-pulse interval)	Unit: ms	
Syllable count	Number of syllables in a single phrase.	Integer	
Phrase intonation	Qualitative description of the pitch contour	Level Rising Falling Rise-Fall Fall-Rise Other Unknown	
Attributes (auditory icon)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Material	Description of the sound source (as reported)		
Location	Where is the signal presented?	Instrument cluster Dashboard Center console HUD Windshield Handheld device Steering wheel Door (s) Combination Other Unknown	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding silence between bursts)	Unit: ms	
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Pulse count	How many pulses are inside the first burst?	Integer	
Pulse duration	Duration of a single pulse (excluding silence between pulses)	Unit: ms	
Inter-pulse interval	Time between pulses (in case a burst contains > 1 pulse)	Unit: ms	
Pulse onset time	Time from silence to maximum volume (i.e., fade-in)	Unit: ms	
Pulse offset time	Time from maximum volume to silence (i.e., fade-out)	Unit: ms	
Pulse intensity	Intensity of the first pulse	Unit: dB dB(A) dB(B) dB(C) dB SPL dBFS	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
		dBV	
		dBu / dBv	
		Knob settings	
		Other	
		Unknown	
Attributes (auditory earcon)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Material	Description of the sound source	Sine / sinusoid / pure tone	
		Beep / tone (unspecified)	
		Musical tone	
		Frequency modulated tone	
		Triangle wave	
		Square wave	
		Sawtooth wave	
		Other	
		Unknown	
Location	Where is the signal presented?	Instrument cluster	
		Dashboard	
		Center console	
		HUD	
		Windshield	
		Handheld device	
		Steering wheel	
		Door	
		(s)	
		Combination	
		Other	
		Unknown	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding silence between bursts)	Unit: ms	
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Pulse count	How many pulses are inside the first burst?	Integer	
Pulse duration	Duration of a single pulse (excluding silence between pulses)	Unit: ms	
Inter-pulse interval	Time between pulses (in case a burst contains > 1 pulse)	Unit: ms	
Pulse onset time	Time from silence to maximum volume (i.e., fade-in)	Unit: ms	
Pulse offset time	Time from maximum volume to silence (i.e., fade-out)	Unit: ms	
Pulse intensity	Intensity of the first pulse	Unit: dB	
		dB(A)	
		dB(B)	
		dB(C)	
		dB SPL	
		dBFS	
		dBV	
		dBu / dBv	
		Knob settings	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
Frequency pulse 1	Frequency (pitch) of the first pulse	Other	
Frequency pulse 2	Frequency (pitch) of the second pulse (if applicable)	Unknown	
Frequency pulse 3	Frequency (pitch) of the third pulse (if applicable)	Unit: Hz	
Frequency pulse 4	Frequency (pitch) of the fourth pulse (if applicable)	Unit: Hz	
Attributes (auditory sonification)		Unit: Hz	
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Material	Description of the sound source	Sine / sinusoid / pure tone	
		Beep / tone (unspecified)	
		Musical tone	
		Frequency modulated tone	
		Triangle wave	
		Square wave	
		Sawtooth wave	
		Other	
		Unknown	
Location	Where is the signal presented?	Instrument cluster	
		Dashboard	
		Center console	
		HUD	
		Windshield	
		Handheld device	
		Steering wheel	
		Door	
		(s)	
		Combination	
		Other	
		Unknown	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding silence between bursts)	Unit: ms	
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Pulse count	How many pulses are inside the first burst?	Integer	
Pulse duration	Duration of a single pulse (excluding silence between pulses)	Unit: ms	
Inter-pulse interval	Time between pulses (in case a burst contains > 1 pulse)	Unit: ms	
Pulse onset time	Time from silence to maximum volume (i.e., fade-in)	Unit: ms	
Pulse offset time	Time from maximum volume to silence (i.e., fade-out)	Unit: ms	
Pulse intensity	Intensity of the first pulse	Unit: dB	
		dB(A)	
		dB(B)	
		dB(C)	
		dB SPL	
		dBFS	
		dBV	
		dBu / dBv	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
		Knob settings	
		Other	
		Unknown	
Frequency pulse 1	Frequency (pitch) of the first pulse	Unit: Hz	
Frequency pulse 2	Frequency (pitch) of the second pulse (if applicable)	Unit: Hz	
Frequency pulse 3	Frequency (pitch) of the third pulse (if applicable)	Unit: Hz	
Frequency pulse 4	Frequency (pitch) of the fourth pulse (if applicable)	Unit: Hz	
Sonification source	Which data stream in- or outside the vehicle modulates the sound?		
Sonification destination	Which sound parameter is being modulated?		
Attributes (tactile vibration)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	
Location	Where is the signal presented?	Steering wheel	
		Chair	
		Torso	
		Wristbands	
		Side stick	
		Pedals	
		Other	
		Unknown	
Burst count	How many bursts are presented? (a burst can have multiple pulses)	Integer	
Burst duration	Duration of a single burst (excluding silence between bursts)	Unit: ms	
Inter-burst interval	Time between bursts (in case of > 1 burst)	Unit: ms	
Pulse count	How many pulses are inside the first burst?	Integer	
Pulse duration	Duration of a single pulse (excluding silence between pulses)	Unit: ms	
Inter-pulse interval	Time between pulses (in case a burst contains > 1 pulse)	Unit: ms	
Pulse onset time	Time from no vibration to maximum vibration (i.e., fade-in)	Unit: ms	
Pulse offset time	Time from maximum vibration to no vibration (i.e., fade-out)	Unit: ms	
Pulse intensity	Intensity of the first pulse	Unit: dB	
		dB(A)	
		dB(B)	
		dB(C)	
		dB SPL	
		dBFS	
		dBV	
		dBu / dBv	
		Knob settings	
		Other	
		Unknown	
Frequency pulse 1	Frequency of the first pulse	Unit: Hz	
Frequency pulse 2	Frequency of the second pulse (if applicable)	Unit: Hz	
Frequency pulse 3	Frequency of the third pulse (if applicable)	Unit: Hz	
Frequency pulse 4	Frequency of the fourth pulse (if applicable)	Unit: Hz	
Size vibration area	Size of the vibration area	Unit: cm2	
Attributes (tactile force feedback)			
Time until takeover	First occurrence of the signal prior to the system limit. Enter the largest duration in case the time window is not fixed, e.g., when takeover time (ToT) is manipulated.	Unit: s	
Presentation duration	How long is the signal presented to the driver?	Unit: ms	

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Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
Location	Where is the signal presented?	Steering wheel Side stick Pedals Other Unknown Unit: %	
Intensity	Proportion of the force applied by the driver		
Background signals			
NDRT manual	Is the driver involved in a manual non-driving related task?	Yes, No, Unknown	
NDRT visual	Is the driver involved in a visual non-driving related task?	Yes, No, Unknown	
NDRT auditory	Is the driver involved in an auditory non-driving related task?	Yes, No, Unknown	
Visual signal presence	Was there a visual non-TOR non-NDRT signal in the cabin in any experimental condition in interval (TOR – 10 s) < t < System limits	Yes, No, Unknown	
Visual signal automation	Is any of the background signals related to the vehicle automation?	Yes, No, Unknown, Not applicable	Use 'Not applicable' in case no visual background signal was present.
Visual spatial proximity	Minimum distance between any TOR signal and any background signal.	Unit: mm	
Visual temporal proximity	Minimum time between any TOR signal and any background signal.	Unit: ms	
Auditory background present	What kind of stationary background noise is present in the cabin? (excluding speech, auditory icons, earcons, sonification, and NDRTs)	No noise Engine noise Road noise Wind noise Music Combination Other Unknown Unit: dB dB(A) dB(B) dB(C) dB SPL dBFS dBV dBu / dBv Knob settings Other Unknown	
Auditory background intensity	Stationary background noise intensity in the cabin	Yes, No, Unknown	
Auditory signal presence	Was there an auditory non-stationary background signal in the cabin in any experimental condition in interval (TOR – 10 s) < t < System limits?	Yes, No, Unknown	
Auditory signal automation	Is any of the auditory non-stationary background signals related to the vehicle automation?	Yes, No, Unknown, Not applicable	
Auditory spatial proximity	Minimum distance between any TOR signal and any non-stationary background signal.	Unit: mm	
Auditory temporal proximity	Minimum time between any TOR signal and any non-stationary background signal.	Unit: ms	
Tactile background intensity	Intensity of non-TOR vibrations in the driver's chair (e.g., road noise)	Unit: dB dB(A) dB(B) dB(C) dB SPL	

(continued on next page)

Table B1 (continued)

Variable	Explanation variable	Categories	Explanation categories
Setting		dBFS dBV dBu / dBv Knob settings Other Unknown	
Traffic density	Number of vehicles in the neighbouring lane(s) during the takeover (select the highest level in case of an experimental manipulation)	Not present 1–10 / km 10–20 / km 20–30 / km > 30 / km Present, # unspecified Manipulated across conditions Presence unknown	Categories taken from Gold et al. (2018) .
SAE before	SAE level before takeover (select the highest level in case of an experimental manipulation)	0 (manual) / 1 (assisted) 2 (partial) 3 (conditional) / 4 (high) Unknown	
SAE after	SAE level after takeover (select the lowest level in case of an experimental manipulation)	0 (manual) / 1 (assisted) 2 (partial) 3 (conditional) / 4 (high) Unknown	
Involvement	Does the vehicle ask the driver to confirm that s/he has taken back the requested control?	Yes, No, Unknown	In Mirnig et al. (2017) , category 'yes' can be interpreted as 'semi-active', and category 'no' can be interpreted as 'passive'.
Completeness of transfer	Full: all functions from 'SAE before' to 'SAE after' are transferred at once; Incremental: step-by-step until transition from 'SAE before' to 'SAE after' is achieved.	Full Incremental Other Unknown	
Fallback	System strategies in case a transition is unsuccessful (select 'no stop' if the driver cannot deny taking over control @ involvement).	Stop in own lane Stop outside own lane No stop (no fallback) Manipulated across conditions Other Unknown	
Input modality	Input modalities available to respond to transition requests.	Physical Touch display Speech Gestures Combination Manipulated across conditions	Category 'physical' includes steering wheel, pedals, buttons, and levers.

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Table B1 (*continued*)

Variable	Explanation variable	Categories	Explanation categories
Mode awareness	Which features are used by the vehicle to inform the driver about the system state before the TOR? (note: this is not about system state changes, but about which features are perceivable before the TOR is presented)	Other Unknown None Symbol Color Text Audio Symbol + Color Symbol + Text Symbol + Audio Symbol + Color + Text Symbol + Text + Audio Symbol + Color + Audio Symbol + Color + Text + Audio Color + Text Color + Audio Color + Text + Audio Text + Audio Other Unknown	
Additional variables TOR reference	Does the paper explicitly refer to another source to describe (any part of) the TOR implementation?	No Yes (appendix) Yes (other paper) Yes (appendix & other paper)	

Appendix C:. Overview of coded studies

See Table C1.

Table C1

Sample description including TOR modality and class. An overview including TOR attributes, background, and setting variables can be found in the digital appendix. V = Visual, A = Auditory, T = Tactile, Y = Yes, N = No, ? = Unknown, - = Not applicable.

Study	Modality	TOR #	Visual text	Visual symbol	Visual abstract	Auditory speech	Auditory icon	Auditory earcon	Tactile vibration
Agrawal et al. (2017)	A	1	–	–	–	Y	N	N	–
Alrefaie et al. (2019)	VA	1	?	?	?	N	N	Y	–
Befelein et al. (2018)	VA	1	Y	Y	N	N	N	Y	–
Befelein et al. (2016)	VA	1	Y	Y	Y	?	?	?	–
	VA	2	Y	Y	Y	?	?	?	–
Berghöfer et al. (2018)	VA	1	Y	Y	N	Y	N	N	–
Blanco et al. (2016)	V	1	Y	N	Y	–	–	–	–
	V	2	Y	N	Y	–	–	–	–
	V	3	Y	N	Y	–	–	–	–
	VT	1	Y	N	Y	–	–	–	Y
	VT	2	Y	N	Y	–	–	–	Y
	VT	3	Y	N	Y	–	–	–	Y
	VAT	1	?	?	?	?	?	?	?
Blommer et al. (2015)	VA	1	?	?	?	?	?	?	–
Blommer et al. (2017)	VA	1	?	?	?	?	?	?	–
Boelhauwer et al. (2020)	VA	1	N	Y	N	Y	N	N	–
Borojeni et al. (2016)	VA	1	N	N	Y	N	N	Y	–
	VA	2	N	N	Y	N	N	Y	–
	VA	3	N	N	Y	N	N	Y	–
Borojeni et al. (2017)	A	1	–	–	–	N	N	Y	–
	AT	1	–	–	–	N	N	Y	Y
Borojeni et al. (2018a)	VA	1	N	N	Y	N	N	Y	–
	VA	2	N	N	Y	N	N	Y	–
	VA	3	N	N	Y	N	N	Y	–
Borojeni et al. (2018b)	VA	1	N	N	Y	N	N	Y	–
	VA	2	N	N	Y	N	N	Y	–
Bourrelly et al. (2019)	VA	1	?	?	?	?	?	?	–
Brandenburg & Chuang (2019)	VA	1	Y	Y	N	N	N	Y	–
	VA	2	Y	Y	N	N	N	Y	–
	VA	3	Y	Y	N	N	N	Y	–
	VA	4	Y	Y	N	N	N	Y	–
Braunagel et al. (2017)	VA	1	?	?	?	?	?	?	–
Bueno et al. (2016)	VA	1	?	?	?	?	?	?	–
Calvi et al. (2020)	A	1	–	–	–	Y	N	Y	–
Cao et al. (2020)	AT	1	–	–	–	Y	N	N	?
	AT	2	–	–	–	Y	N	N	?
Çapalar & Olaverri-Monreal (2017)	A	1	–	–	–	?	?	?	–
Chauvin et al. (2020)	VA	1	?	?	?	?	?	?	–
Chen & Chen (2021)	A	1	–	–	–	Y	N	N	–
Chen et al. (2020)	A	1	–	–	–	N	N	Y	–
	A	2	–	–	–	N	N	Y	–
	A	3	–	–	–	N	N	Y	–
	A	4	–	–	–	N	N	Y	–
Choi et al. (2020)	VA	1	N	Y	N	Y	N	N	–
Clark & Feng (2017)	A	1	–	–	–	N	N	Y	–
	A	2	–	–	–	N	N	Y	–
Clark et al. (2019)	VA	1	?	?	?	Y	?	?	–
	VAT	1	N	Y	N	Y	?	?	Y
Cohen-Lazry et al. (2018)	T	1	–	–	–	–	–	–	Y
	T	2	–	–	–	–	–	–	Y
Cohen-Lazry et al. (2017)	A	1	–	–	–	Y	N	N	–
Cohen-Lazry et al. (2019)	T	1	–	–	–	–	–	–	Y
	T	2	–	–	–	–	–	–	Y
Cohen-Lazry et al. (2020)	A	1	–	–	–	N	N	Y	–
Cui et al. (2018)	V	1	Y	N	N	–	–	–	–
	V	2	Y	N	N	–	–	–	–
	V	3	Y	N	N	–	–	–	–
de Winter et al. (2016)	V	1	Y	?	N	–	–	–	–
	VA	1	N	Y	N	N	N	Y	–

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Table C1 (continued)

Study	Modality	TOR #	Visual text	Visual symbol	Visual abstract	Auditory speech	Auditory icon	Auditory earcon	Tactile vibration
DinparastDjadid et al. (2019)	VA	2	N	Y	N	N	N	Y	–
	VA	3	N	Y	N	N	N	Y	–
Dogan et al. (2017)	VA	1	Y	N	N	N	N	Y	–
Dogan et al. (2019)	VA	1	Y	Y	N	?	?	?	–
Doubek et al. (2020)	VA	1	Y	Y	N	N	N	Y	–
	VA	2	Y	Y	N	N	N	Y	–
	VA	3	Y	Y	N	N	N	Y	–
Du et al. (2020a)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	N	–
Du et al. (2020b)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	N	–
Duthoit et al. (2018)	T	1	–	–	–	–	–	–	Y
	T	2	–	–	–	–	–	–	Y
	T	3	–	–	–	–	–	–	Y
	T	4	–	–	–	–	–	–	Y
Ebnali et al. (2019)	–	1	–	–	–	–	–	–	–
Eriksson & Stanton (2017a)	VA	1	Y	Y	N	Y	N	N	–
Eriksson & Stanton (2017b)	VA	1	Y	Y	N	Y	N	N	–
Eriksson et al. (2017)	VA	1	N	Y	N	Y	N	N	–
Eriksson et al. (2019)	T	1	–	–	–	–	–	–	Y
	VT	1	N	N	Y	–	–	–	Y
	VT	2	N	N	Y	–	–	–	Y
	VT	3	N	Y	N	–	–	–	Y
Favarò et al. (2019)	VA	1	N	Y	N	Y	N	N	–
Feldhütter et al. (2017a)	A	1	–	–	–	?	?	?	–
Feldhütter et al. (2017b)	VA	1	Y	Y	N	N	N	Y	–
Feldhütter et al. (2018)	VA	1	?	?	?	N	N	Y	–
Feldhütter et al. (2019)	A	1	–	–	–	N	N	Y	–
Fleskes & Hurwitz (2019)	VA	1	Y	Y	N	Y	N	Y	–
	VA	2	Y	Y	N	Y	N	Y	–
	VA	3	Y	Y	N	Y	N	Y	–
Forster et al. (2016)	V	1	Y	Y	Y	–	–	–	–
	V	2	Y	Y	Y	–	–	–	–
	V	3	Y	Y	Y	–	–	–	–
Forster et al. (2017)	V	1	Y	Y	N	–	–	–	–
	VA	1	Y	Y	N	N	N	Y	–
Funkhouser & Drews (2016)	A	1	–	–	–	N	N	Y	–
Geitner et al. (2019)	A	1	–	–	–	N	N	Y	–
	T	1	–	–	–	–	–	–	Y
	AT	1	–	–	–	N	N	Y	Y
Gold et al. (2015a)	VA	1	N	Y	N	Y	N	N	–
	VA	2	N	Y	N	Y	N	N	–
	VA	3	N	Y	N	Y	N	N	–
Gold et al. (2015b)	VA	1	N	Y	N	N	N	Y	–
Gold et al. (2016)	A	1	–	–	–	N	N	Y	–
Hayashi et al. (2021)	VA	1	Y	N	N	N	N	Y	–
	VA	2	Y	N	N	N	N	Y	–
He et al. (2021)	VA	1	Y	Y	N	N	N	Y	–
	VA	2	?	?	?	?	?	?	–
Hecht et al. (2020)	VA	1	Y	N	Y	N	N	Y	–
	VA	2	Y	N	Y	N	N	Y	–
Hergeth et al. (2016)	VA	1	N	Y	N	N	N	Y	–
Hester et al. (2017)	A	1	–	–	–	N	N	Y	–
	A	2	–	–	–	Y	N	N	–
Ito et al. (2016)	A	1	–	–	–	N	N	Y	–
	A	2	–	–	–	N	N	Y	–
Jarosch et al. (2017)	–	1	–	–	–	–	–	–	–
Jarosch et al. (2019a)	VA	1	Y	Y	N	?	?	?	–
Jarosch et al. (2019b)	VA	1	Y	N	N	?	?	?	–
	VA	2	Y	N	N	?	?	?	–
Jiménez et al. (2018)	A	1	–	–	–	N	N	Y	–
Kamezaki et al. (2020)	VA	1	N	Y	Y	N	N	Y	–
Kerschbaum et al. (2015)	VA	1	N	Y	N	N	N	Y	–
Kerschbaum et al. (2017)	VA	1	N	Y	Y	N	N	Y	–
Kim & Yang (2017)	VA	1	Y	N	N	N	N	Y	–

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Table C1 (continued)

Study	Modality	TOR #	Visual text	Visual symbol	Visual abstract	Auditory speech	Auditory icon	Auditory earcon	Tactile vibration
Kim & Yang (2020)	V	1	Y	Y	Y	–	–	–	–
	A	1	–	–	–	N	N	Y	–
	T	1	–	–	–	–	–	–	Y
	VA	1	Y	Y	Y	N	N	Y	–
	VA	2	Y	Y	Y	N	N	Y	–
	VT	1	Y	Y	Y	–	–	–	Y
	AT	1	–	–	–	N	N	Y	Y
	VAT	1	Y	Y	Y	N	N	Y	Y
Kim et al. (2017)	A	1	–	–	–	Y	N	N	–
	VA	1	Y	N	N	N	N	Y	–
Kim et al. (2021)	A	1	–	–	–	Y	N	N	–
Ko & Ji (2018)	A	1	–	–	–	?	?	?	–
Körber et al. (2016a)	A	1	–	–	–	N	N	Y	–
Körber et al. (2016b)	–	1	–	–	–	–	–	–	–
Körber et al. (2018)	A	1	–	–	–	N	N	Y	–
Kraus et al. (2019)	V	1	Y	N	Y	–	–	–	–
Langlois & Soualmi (2016)	VA	1	Y	Y	N	?	?	?	–
Lapoehn et al. (2015)	V	1	Y	Y	Y	–	–	–	–
	VA	1	Y	Y	Y	?	?	?	–
Lapoehn et al. (2016)	A	1	–	–	–	?	?	?	–
	VA	1	Y	Y	N	?	?	?	–
Lau et al. (2018)	VA	1	N	Y	N	N	N	Y	–
	VA	2	Y	N	Y	N	N	Y	–
Lee & Yang (2020)	V	1	?	?	?	–	–	–	–
	A	1	–	–	–	?	?	?	–
	T	1	–	–	–	–	–	–	?
	VA	1	?	?	?	?	?	?	–
	VT	1	?	?	?	–	–	–	?
	AT	1	–	–	–	?	?	?	?
	VAT	1	?	?	?	?	?	?	?
Lee et al. (2021)	–	1	–	–	–	–	–	–	–
Li et al. (2018)	VA	1	Y	N	N	Y	N	N	–
Li et al. (2019a)	VA	1	Y	N	N	Y	N	N	–
Li et al. (2019b)	VA	1	Y	N	N	Y	N	N	–
	VA	2	Y	N	N	Y	N	N	–
Li et al. (2020)	VA	1	Y	N	N	N	N	Y	–
Lin et al. (2020)	VA	1	Y	Y	N	Y	N	N	–
	VA	2	Y	Y	N	Y	N	N	–
	VA	3	Y	Y	N	Y	N	N	–
Lorenz et al. (2014)	VA	1	N	Y	N	N	N	Y	–
	VA	2	N	Y	Y	N	N	Y	–
	VA	3	N	Y	Y	N	N	Y	–
Louw et al. (2015)	A	1	–	–	–	N	N	Y	–
Louw et al. (2021)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	N	N	Y	–
	A	3	–	–	–	N	N	Y	–
Lu et al. (2019)	VA	1	Y	Y	N	Y	N	Y	–
Maggi et al. (2020)	A	1	–	–	–	Y	N	N	–
Manawadu et al. (2018)	VA	1	N	Y	Y	N	N	Y	–
McNabb (2017)	A	1	–	–	–	N	N	Y	–
	A	2	–	–	–	N	N	Y	–
	A	3	–	–	–	N	N	Y	–
	A	4	–	–	–	N	N	Y	–
Melcher et al. (2015)	VA	1	Y	Y	N	N	N	Y	–
	VA	2	Y	Y	N	N	N	Y	–
Miller et al. (2015)	VA	1	Y	N	N	N	N	Y	–
Miller et al. (2016)	VA	1	N	Y	N	Y	N	N	–
Mok et al. (2015a)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	N	–
	A	3	–	–	–	Y	N	N	–
Mok et al. (2015b)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	N	–
	A	3	–	–	–	Y	N	N	–
Mok et al. (2017)	VA	1	?	?	?	Y	N	N	–
	VA	2	?	?	?	Y	N	N	–
	VA	3	?	?	?	Y	N	N	–
Molnar (2017)	VAT	1	N	N	Y	Y	N	N	Y
Morgan et al. (2018)	A	1	–	–	–	?	?	?	–

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Table C1 (continued)

Study	Modality	TOR #	Visual text	Visual symbol	Visual abstract	Auditory speech	Auditory icon	Auditory earcon	Tactile vibration
Müller et al. (2021)	VAT	1	N	Y	N	N	N	Y	Y
Naujoks et al. (2014)	V	1	N	Y	N	–	–	–	–
	VA	1	N	Y	N	N	N	Y	–
Naujoks et al. (2015)	VA	1	N	Y	N	N	N	Y	–
	VA	2	Y	Y	N	Y	N	N	–
Naujoks et al. (2017a)	VA	1	Y	Y	N	N	N	Y	–
Naujoks et al. (2017b)	V	1	Y	Y	Y	–	–	–	–
	V	2	Y	Y	Y	–	–	–	–
	V	3	Y	Y	Y	–	–	–	–
Naujoks et al. (2018)	VA	1	Y	Y	Y	?	?	?	–
Naujoks et al. (2019a)	VA	1	Y	Y	Y	Y	N	N	–
Nilsson et al. (2015)	A	1	–	–	–	?	?	?	–
Pampel et al. (2019)	V	1	?	?	?	–	–	–	–
Payre et al. (2015)	VA	1	Y	N	N	Y	?	?	–
Payre et al. (2017)	VA	1	Y	N	N	Y	N	Y	–
Petermeijer et al. (2017a)	A	1	–	–	–	N	N	Y	–
	A	2	–	–	–	N	N	Y	–
	T	1	–	–	–	–	–	–	Y
	T	2	–	–	–	–	–	–	Y
	AT	1	–	–	–	N	N	Y	Y
	AT	2	–	–	–	N	N	Y	Y
Petermeijer et al. (2017b)	T	1	–	–	–	–	–	–	Y
	T	2	–	–	–	–	–	–	Y
Petermeijer et al. (2017c)	V	1	N	N	Y	–	–	–	–
	A	1	–	–	–	N	N	Y	–
	T	1	–	–	–	–	–	–	Y
Politis et al. (2015)	V	1	Y	N	N	–	–	–	–
	V	2	Y	N	N	–	–	–	–
	V	3	Y	N	N	–	–	–	–
	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	N	–
	A	3	–	–	–	Y	N	N	–
	T	1	–	–	–	–	–	–	–
	T	2	–	–	–	–	–	–	–
	T	3	–	–	–	–	–	–	–
	VA	1	Y	N	N	Y	N	N	–
	VA	2	Y	N	N	Y	N	N	–
	VA	3	Y	N	N	Y	N	N	–
	VT	1	Y	N	N	–	–	–	–
	VT	2	Y	N	N	–	–	–	–
	VT	3	Y	N	N	–	–	–	–
	AT	1	–	–	–	Y	N	N	–
	AT	2	–	–	–	Y	N	N	–
	AT	3	–	–	–	Y	N	N	–
	VAT	1	Y	N	N	Y	N	N	–
	VAT	2	Y	N	N	Y	N	N	–
	VAT	3	Y	N	N	Y	N	N	–
Politis et al. (2017)	V	1	N	N	Y	–	–	–	–
	V	2	Y	N	N	–	–	–	–
	V	3	N	N	Y	–	–	–	–
	V	4	Y	N	N	–	–	–	–
	A	1	–	–	–	N	N	Y	–
	A	2	–	–	–	Y	N	N	–
	A	3	–	–	–	N	N	Y	–
	A	4	–	–	–	Y	N	N	–
	T	1	–	–	–	–	–	–	Y
	T	2	–	–	–	–	–	–	Y
	T	3	–	–	–	–	–	–	Y
	T	4	–	–	–	–	–	–	Y
	VA	1	N	N	Y	N	N	Y	–
	VA	2	Y	N	N	Y	N	N	–
	VA	3	N	N	Y	N	N	Y	–
	VA	4	Y	N	N	Y	N	N	–
	VT	1	N	N	Y	–	–	–	Y
	VT	2	Y	N	N	–	–	–	Y
	VT	3	N	N	Y	–	–	–	Y
	VT	4	Y	N	N	–	–	–	Y
	AT	1	–	–	–	N	N	Y	Y

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Table C1 (continued)

Study	Modality	TOR #	Visual text	Visual symbol	Visual abstract	Auditory speech	Auditory icon	Auditory earcon	Tactile vibration
	AT	2	–	–	–	Y	N	N	Y
	AT	3	–	–	–	N	N	Y	Y
	AT	4	–	–	–	Y	N	N	Y
	VAT	1	N	N	Y	N	N	Y	Y
	VAT	2	Y	N	N	Y	N	N	Y
	VAT	3	N	N	Y	N	N	Y	Y
	VAT	4	Y	N	N	Y	N	N	Y
Politis et al. (2018)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	N	–
	VAT	1	N	N	Y	Y	N	N	Y
Radlmayr et al. (2014)	VA	1	N	Y	N	N	N	Y	–
Radlmayr et al. (2018)	VA	1	N	Y	N	N	N	Y	–
Radlmayr et al. (2019)	VA	1	?	?	?	N	N	Y	–
Rauffet et al. (2020)	VA	1	?	?	?	?	?	?	–
Roche & Brandenburg (2018)	AT	1	–	–	–	Y	N	N	Y
	AT	2	–	–	–	Y	N	N	Y
	AT	3	–	–	–	Y	N	N	Y
	AT	4	–	–	–	Y	N	N	Y
Roche et al. (2018)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	N	–
	VA	1	Y	N	N	N	N	Y	–
	VA	2	Y	N	N	N	N	Y	–
Roche et al. (2020)	A	1	–	–	–	N	N	Y	–
	A	2	–	–	–	N	N	Y	–
	A	3	–	–	–	N	N	Y	–
Ruscio et al. (2017)	V	1	N	N	Y	–	–	–	–
Scharfe et al. (2020)	–	1	–	–	–	–	–	–	–
Schartmüller et al. (2018a)	VA	1	Y	N	N	?	?	?	–
Schartmüller et al. (2018b)	VA	1	Y	N	N	N	N	Y	–
Schmidt et al. (2017)	VA	1	N	Y	N	N	N	Y	–
Schnebelen et al. (2020)	VA	1	Y	Y	N	N	N	Y	–
Schwalk et al. (2015)	T	1	–	–	–	–	–	–	Y
Schwarz et al. (2019)	VA	1	Y	N	N	N	N	Y	–
	VA	2	Y	N	N	N	N	Y	–
	VA	3	Y	N	N	N	N	Y	–
Sportillo et al. (2017)	VA	1	Y	N	N	N	N	Y	–
	VA	2	Y	N	N	N	N	Y	–
Sportillo et al. (2018)	V	2	Y	N	N	–	–	–	–
	V	4	Y	N	N	–	–	–	–
	VA	1	Y	Y	N	N	N	Y	–
	VA	3	Y	Y	N	N	N	Y	–
Stanton et al. (2022)	A	1	–	–	–	Y	N	N	–
	VA	1	N	Y	N	Y	N	N	–
Sun et al. (2021)	VAT	1	N	Y	N	Y	N	N	?
Talamonti (2017)	VA	1	?	?	?	Y	?	?	–
	VA	2	?	?	?	Y	?	?	–
	VA	3	?	?	?	Y	?	?	–
	VA	4	?	?	?	Y	?	?	–
Tang et al. (2020)	A	1	–	–	–	N	N	Y	–
	T	1	–	–	–	–	–	–	Y
	AT	1	–	–	–	N	N	Y	Y
Telpaz et al. (2015)	VA	1	Y	N	N	N	N	Y	–
Telpaz et al. (2017)	VA	1	Y	N	N	N	N	Y	–
van den Beukel et al. (2016)	A	1	–	–	–	N	Y	N	–
	VA	1	N	Y	N	N	Y	N	–
	VAT	1	N	N	Y	N	Y	N	Y
van der Heiden et al. (2017)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	Y	–
	A	3	–	–	–	Y	N	Y	–
van der Meulen et al. (2016)	VA	1	Y	Y	N	N	N	Y	–
Várhelyi et al. (2021)	VA	1	N	N	Y	Y	N	N	–
Vlakveld et al. (2018)	VA	1	Y	N	N	N	N	Y	–
	VA	2	Y	N	N	N	N	Y	–
Vogelpohl et al. (2018)	VA	1	N	Y	N	N	N	Y	–
Vogelpohl et al. (2019)	VA	1	N	Y	N	?	?	?	–
Vogelpohl et al. (2020)	VA	1	N	Y	N	N	N	Y	–
Walch et al. (2016)	VA	1	Y	Y	N	Y	N	N	–

(continued on next page)

Table C1 (continued)

Study	Modality	TOR #	Visual text	Visual symbol	Visual abstract	Auditory speech	Auditory icon	Auditory earcon	Tactile vibration
Walch et al. (2017)	VA	1	Y	Y	N	Y	N	Y	–
	VA	2	Y	Y	N	Y	N	Y	–
	VA	3	Y	Y	N	Y	N	Y	–
Wan & Wu (2018a)	VAT	1	N	N	Y	Y	N	N	Y
Wan & Wu (2018b)	T	1	–	–	–	–	–	–	Y
Wandtner (2018)	VA	1	N	?	?	?	?	?	–
	VA	2	?	?	?	N	?	?	–
	VA	3	N	?	?	N	?	?	–
Wandtner et al. (2018a)	VA	1	N	Y	N	N	N	Y	–
	VA	2	N	Y	N	N	N	Y	–
Wandtner et al. (2018b)	VA	1	N	Y	N	?	?	?	–
Wandtner et al. (2018c)	VA	1	N	?	?	?	?	?	–
Wang & Söffker (2019)	VA	1	?	?	?	Y	N	N	–
Weidner & Broll (2020)	VA	1	N	Y	N	N	N	Y	–
	VA	2	N	Y	N	N	N	Y	–
	VA	3	N	Y	N	N	N	Y	–
	VA	4	N	Y	N	N	N	Y	–
Weinbeer et al. (2017)	VA	1	Y	N	N	?	?	?	–
Wiedemann et al. (2018)	VA	1	Y	Y	N	Y	N	Y	–
Wiegand et al. (2018)	A	1	–	–	–	N	N	Y	–
Wintersberger et al. (2018)	V	1	Y	Y	N	–	–	–	–
	V	2	Y	Y	N	–	–	–	–
	V	3	Y	Y	N	–	–	–	–
	V	4	Y	Y	N	–	–	–	–
Wintersberger et al. (2019)	V	1	Y	N	N	–	–	–	–
	V	2	Y	N	N	–	–	–	–
Wörle et al. (2020)	VA	1	N	N	Y	Y	N	N	–
	VA	2	N	N	Y	Y	N	N	–
	VA	3	N	N	Y	N	N	Y	–
	VA	4	N	N	Y	N	N	Y	–
Wright et al. (2017)	A	1	–	–	–	Y	?	?	–
Wright et al. (2018)	A	1	–	–	–	Y	N	N	–
Wu et al. (2019a)	VA	1	N	Y	N	Y	N	N	–
Wu et al. (2019b)	VA	1	Y	Y	N	Y	N	N	–
Wu et al. (2020)	VA	1	N	Y	N	Y	N	N	–
Wulf et al. (2015)	A	1	–	–	–	?	?	?	–
Yang et al. (2017)	V	1	N	N	Y	–	–	–	–
Yang et al. (2018a)	VA	1	N	Y	Y	N	N	Y	–
Yang et al. (2018b)	VA	1	N	Y	N	N	N	Y	–
Yang et al. (2020)	A	1	–	–	–	N	N	Y	–
Yoon & Ji (2019)	A	1	–	–	–	N	N	Y	–
Yoon et al. (2019)	V	1	N	Y	N	–	–	–	–
	A	1	–	–	–	N	N	Y	–
	T	1	–	–	–	–	–	–	Y
	VA	1	N	Y	N	N	N	Y	–
	VT	1	N	Y	N	–	–	–	Y
	AT	1	–	–	–	N	N	Y	Y
	VAT	1	N	Y	N	N	N	Y	Y
	A	1	–	–	–	N	N	Y	–
Yoon et al. (2021)	V	1	Y	Y	N	–	–	–	–
Yun & Yang (2020)	A	1	–	–	–	N	N	Y	–
	T	1	–	–	–	–	–	–	Y
	VA	1	Y	Y	N	N	N	Y	–
	VT	1	Y	Y	N	–	–	–	Y
	AT	1	–	–	–	N	N	Y	Y
	VAT	1	Y	Y	N	N	N	Y	Y
	VA	1	N	Y	N	N	N	Y	–
	VA	1	N	Y	N	N	N	Y	–
Zeeb et al. (2015)	VA	1	N	Y	N	N	N	N	–
Zeeb et al. (2016)	VA	1	N	Y	N	N	N	Y	–
Zeeb et al. (2017)	VA	1	Y	Y	N	Y	N	N	–
Zhang et al. (2021)	A	1	–	–	–	Y	N	N	–
	A	2	–	–	–	Y	N	N	–
	A	3	–	–	–	Y	N	N	–
	A	4	–	–	–	Y	N	N	–
Zhou et al. (2020)	VA	1	?	?	?	?	?	?	–

Appendix D:. Interrater reliability

In general there are two approaches to assess whether coders have the same interpretation of a codebook. In one approach, all materials are processed by at least two coders, after which disagreements between coders are resolved (e.g., [Mirnig et al., 2017](#); [Dey et al., 2020](#)). Measures of agreement are typically not calculated. In another approach, a sample of the materials is processed by multiple coders (e.g., [Zhang et al., 2019](#)), after which measures of agreement may be calculated. This second approach was chosen in the present study, due to the relatively large sample size and number of variables, as well as the opportunity to gain insights into the clarity of the operational definitions of the codebook.

The agreement between the coders was assessed in three rounds using a random sample of 5 studies per round of testing (without duplicates of formerly coded studies). Scores resulting from Round 1 and Round 2 were used to facilitate plenary discussions on understanding causes for disagreement, occasionally leading to refinements of definitions and operationalizations in the codebook. Individual bulk coding of the studies started after Round 2. Finally, a third interrater reliability test round (Round 3) was performed to probe whether the joint interpretation of the codebook was constant and sufficient throughout the coding process.

Percentage agreement for each variable was calculated as the proportion of papers at which all coders agreed. The average percentage agreement was then calculated across variables belonging to the same group (i.e., TOR Modality, TOR Class, TOR Attribute, Background signals, and Setting). Starting at the highest hierarchical level for TOR signals, for each modality (e.g., Visual), disagreement across the coders on whether a modality was used in a particular paper resulted in exclusion of that paper from further analysis at the level of TOR Class (e.g., Text, Symbol, Abstract, Anthropomorphic), because missing values would be introduced otherwise. Likewise, if coders disagreed on the class(es) used in a particular paper, then that paper was not further analyzed at the level of TOR Attributes for the classes at which disagreement was found. Previous studies have used a criterion of 80 % above which a variable is said to have an adequate reliability to be considered for subsequent analysis using statistical models (e.g., [Varotto et al., 2021](#)). In our study, we have used the 80 % threshold as an indicator for adequate reliability, but seeing that no further statistical models were constructed, we have chosen to describe all variables, including variables with a percentage agreement below 80 %.

For each group of variables the average agreement across coders is depicted in [Fig. D.1](#). For TOR signals, the agreement was highest for Modality, followed by Class, and lowest for Attribute. This order suggests that coding a paper becomes more difficult the more one zooms in. In Round 2, i.e., prior to individually coding the bulk of the papers, the average agreement was above 80 % for all groups, except Background signals. By the end of the coding process, the average agreement showed a decrease in all groups, with an average agreement below 80 % for Attribute, Setting, and Background signals.

Within Attributes, variables relating to the number of bursts and pulses most often yielded disagreement between coders in Round 3 (but not in Round 2). Such disagreement typically concerned whether the number of bursts or pulses was known or unknown.

For Setting, underlying variables with an agreement below 80 % during Round 2 and/or Round 3 were Traffic density (both rounds), Fallback (one round), Involvement (both rounds), and Mode awareness indication (both rounds). The percentage agreement on Fallback and Mode awareness (and also their deducibility, see below) was surprisingly low, considering that the operational definitions were copied from [Mirnig et al. \(2017\)](#). In that study, all papers were coded by two researchers separately, after which disagreements were corrected. [Mirnig et al. \(2017\)](#) do not report how often disagreement occurred, which means it is not possible to make a comparison with the percentage agreement found in the present study. Nonetheless, our findings indicate that Fallback and Mode awareness may require a clearer definition.

Background signals showed an agreement below 80 % in Round 2 on the underlying variables specifying the presence of visual and auditory background signals and whether auditory background noise was present. By Round 3 this was resolved, except for the presence of visual background signals. Furthermore, coders disagreed on all types of NDRT-involvement (e.g., manual, visual, auditory; here interpreted as background signals due to potential masking of and distraction from TORs) in Round 3.

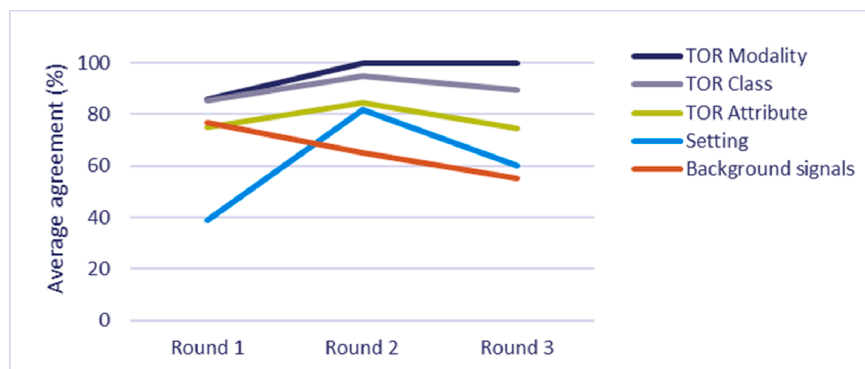


Fig. D1. Agreement for three coding rounds, averaged across coders and variables within each variable group.

Appendix E.: Implementation of attributes

See Figs. E1-E6.

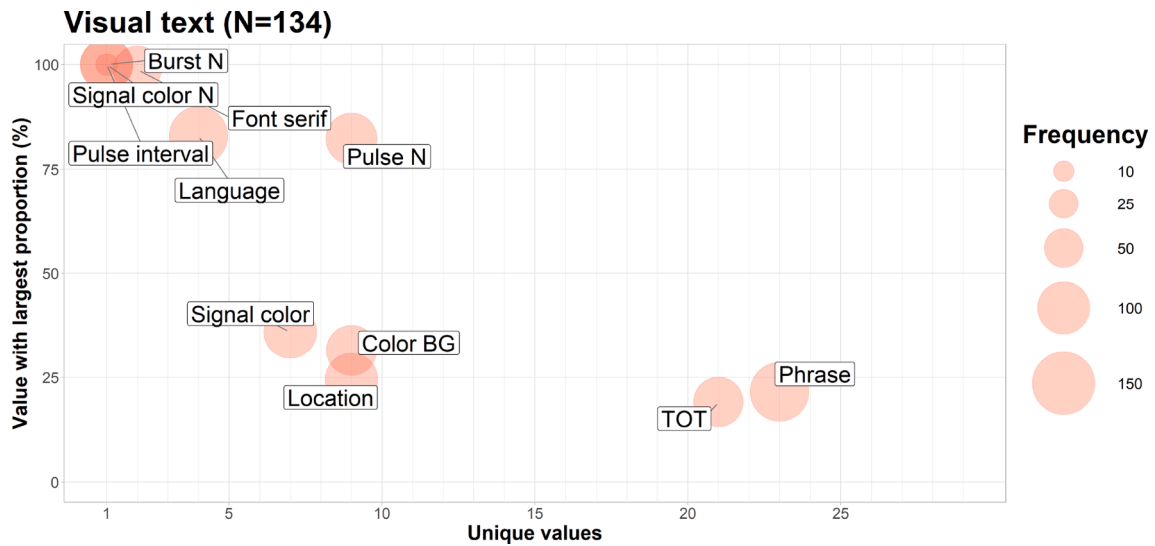


Fig. E1. Number of unique values and proportion of the largest value within attributes of the visual text class. Only attributes with a deducibility of at least 50 % are depicted. 'N' = count, 'BG' = background, 'TOT' = takeover time.

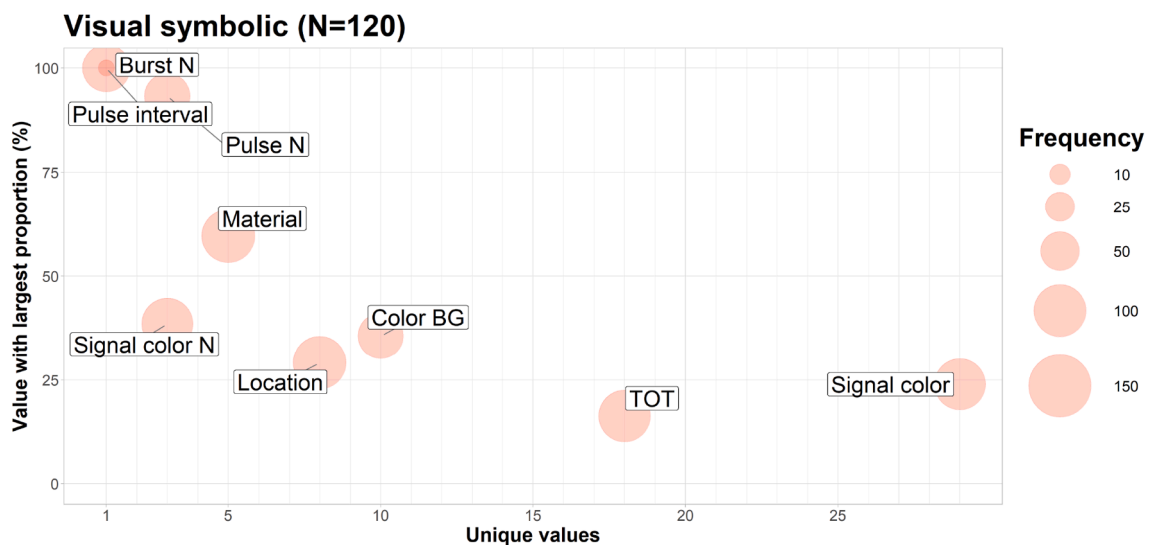


Fig. E2. Number of unique values and proportion of the largest value within attributes of the visual symbolic class. Only attributes with a deducibility of at least 50 % are depicted. 'N' = count, 'BG' = background, 'TOT' = takeover time.

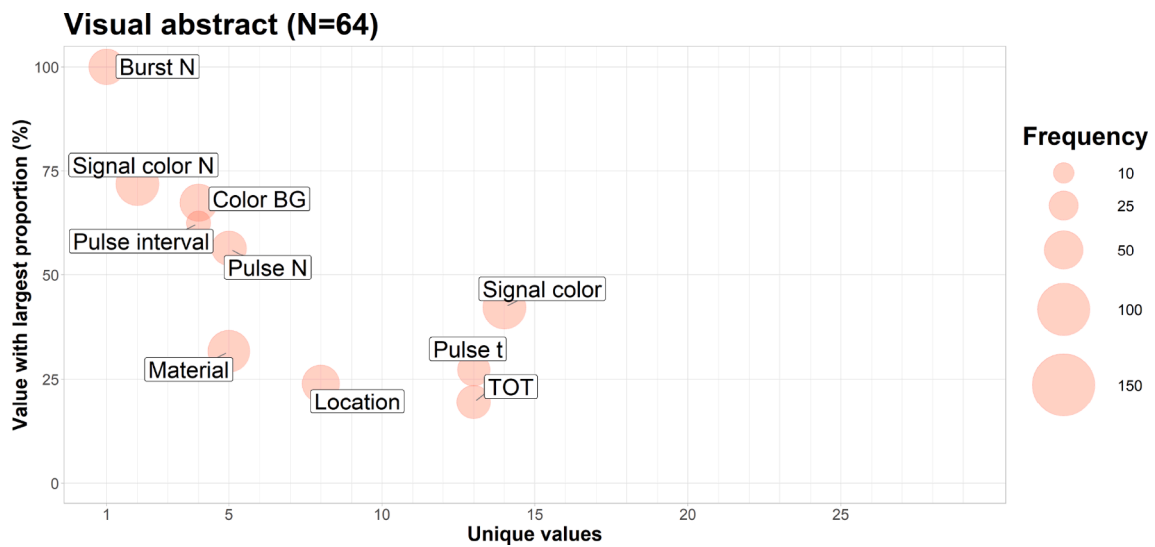


Fig. E3. Number of unique values and proportion of the largest value within attributes of the visual abstract class. Only attributes with a deducibility of at least 50 % are depicted. 'N' = count, 'BG' = background, 'TOT' = takeover time.

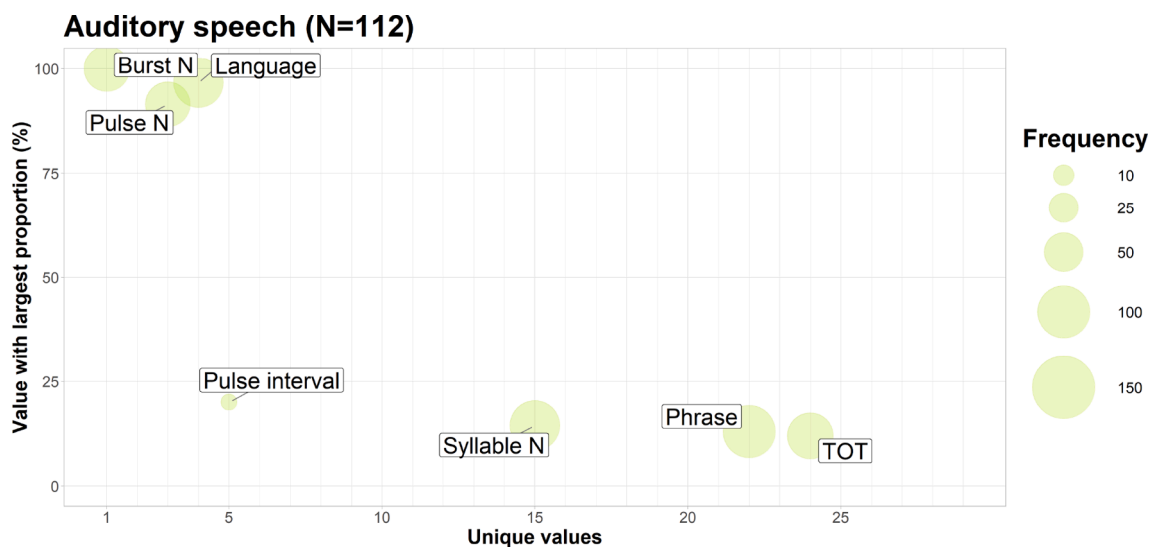


Fig. E4. Number of unique values and proportion of the largest value within attributes of the auditory speech class. Only attributes with a deducibility of at least 50 % are depicted. 'N' = count, 'TOT' = takeover time.

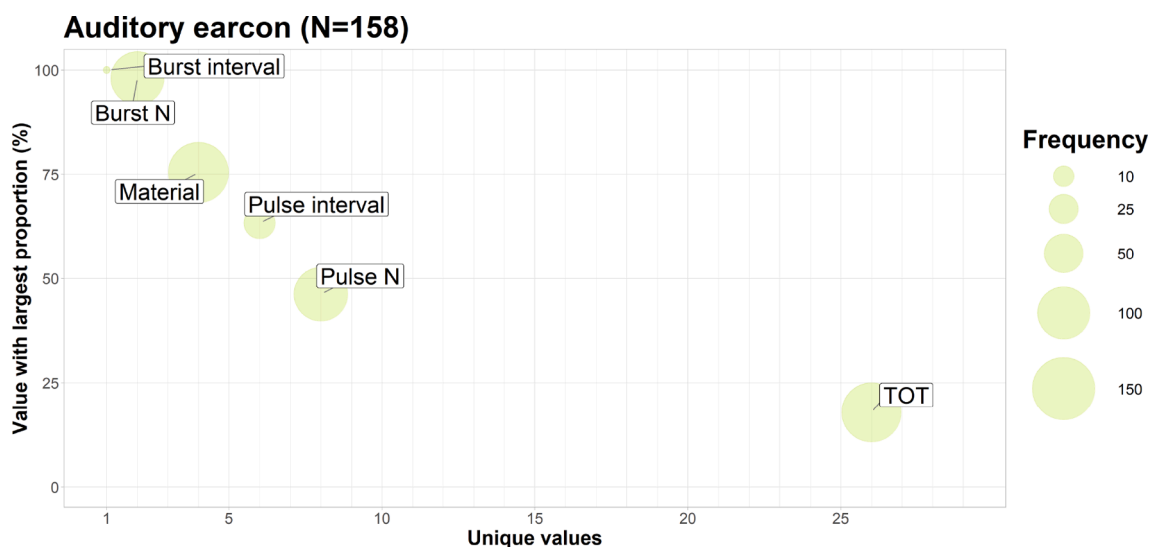


Fig. E5. Number of unique values and proportion of the largest value within attributes of the auditory earcon class. Only attributes with a deducibility of at least 50 % are depicted. 'N' = count, 'TOT' = takeover time.

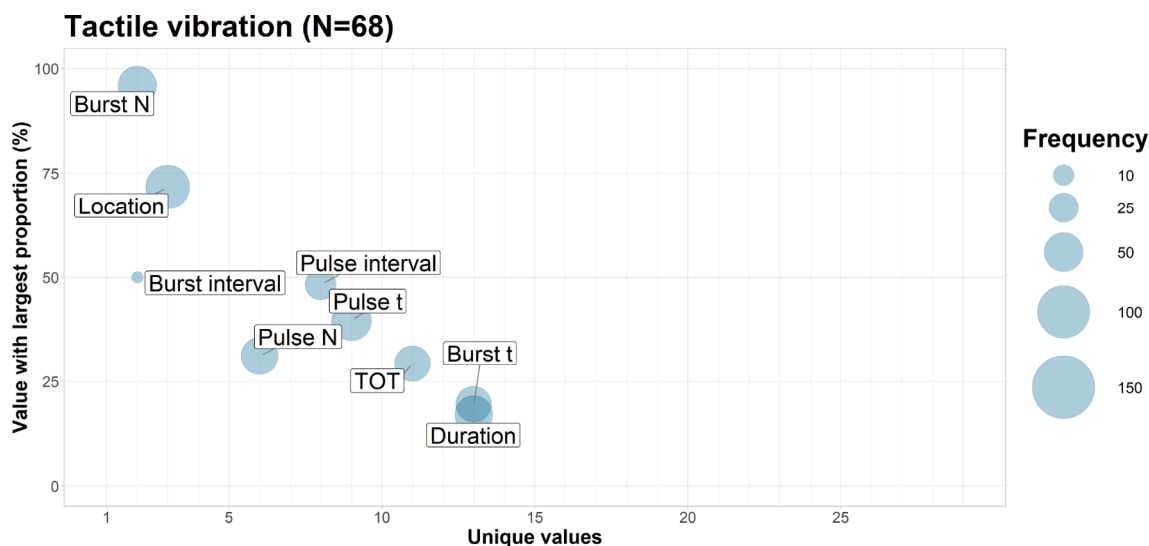


Fig. E6. Number of unique values and proportion of the largest value within attributes of the tactile vibration class. Only attributes with a deducibility of at least 50 % are depicted. 'N' = count, 't' = burst or pulse duration. 'TOT' = takeover time.

Appendix F. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.trf.2022.10.009>.

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*References marked with an asterisk indicate studies included in the systematic review.

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